Statistics Class 11 - Solved Exercises (PCTB, English medium)

Complete book: Exercises 1-9, solved from the 2017-18 10th edition textbook (May 2017 printing)

Chapter 1: Introduction to Statistics

Exercise 1 - Questions 1.1 to 1.8

1.1 Define the word statistics and explain its different meanings?

The word 'statistics' is used in two main senses. As a plural noun it means numerical facts or data systematically collected, e.g. the statistics of births, prices or rainfall. As a singular noun it means the science (the body of methods) used for collecting, organising, presenting, analysing and interpreting numerical data and for drawing conclusions from them. The book notes four present-day uses of the word: statistics as numerical data, as a science of methods, as the results of applying those methods, and as estimates or computed values.

1.2 Define the following terms: (i) Population and sample (ii) Parameter and statistic.

Population: the total set of all items or individuals about which information is required, e.g. all students of a college. Sample: a part (a subset) of the population selected for study, e.g. 50 students chosen from the college. Parameter: a numerical quantity computed from the population, e.g. the mean height of all students. Statistic: a numerical quantity computed from the sample, e.g. the mean height of the 50 selected students; it is used to estimate the parameter.

1.3 Distinguish between qualitative and quantitative variables?

Qualitative (categorical) variables describe a quality or attribute that cannot be measured in numbers, e.g. sex, religion, colour, level of satisfaction; they are

counted in categories. Quantitative variables take numerical values that can be measured or counted, e.g. height, weight, marks, income. Quantitative variables are further divided into discrete (countable, e.g. number of children) and continuous (measurable on a scale, e.g. height).

1.4 (i) Write the following using a summation sign with appropriate index: (a) Y_3 + Y_4 + ... + Y_15 (b) Y_1 squared + Y_2 squared + Y_3 squared + Y_4 squared (c) (Y_1 - mu) squared + (Y_2 - mu) squared + (Y_3 - mu) squared (d) bY_20 + bY_21 + ... + bY_30. (ii) Expand the following summation and product signs: (a) sum i = 1 to 5 Y_i (b) sum i = 5 to 8 (Y_i - mu) (c) sum i = 1 to 3 Y_i squared (d) sum i = 2 to 4 (Y_i - 9) (e) sum i = 1 to 3 Y_i (f) product i = 3 to 6 (aY_i) squared (g) product i = 2 to 4 (X_i - Y_i).

(i) Using a summation sign with appropriate index: (a) sum from i = 3 to 15 of Y_i. (b) sum from i = 1 to 4 of Y_i squared. (c) sum from i = 1 to 3 of (Y_i - mu) squared. (d) b times sum from i = 20 to 30 of Y_i. (ii) Expansions: (a) Y_1 + Y_2 + Y_3 + Y_4 + Y_5. (b) (Y_5 - mu) + (Y_6 - mu) + (Y_7 - mu) + (Y_8 - mu). (c) Y_1 squared + Y_2 squared + Y_3 squared. (d) (Y_2 - 9) + (Y_3 - 9) + (Y_4 - 9). (e) Y_1 + Y_2 + Y_3. (f) (aY_3) squared times (aY_4) squared times (aY_5) squared times (aY_6) squared. (g) (X_2 - Y_2) times (X_3 - Y_3) times (X_4 - Y_4).

1.5 Classify the following as categorical, discrete or continuous variable: (i) Sex of an insect (ii) Weights of plants (iii) Major crops of Pakistan (iv) Level of satisfaction (v) Teaching standards (vi) Temperature measured in Fahrenheit.

(i) Sex of an insect: categorical (qualitative). (ii) Weights of plants: continuous (quantitative). (iii) Major crops of Pakistan: categorical (qualitative). (iv) Level of satisfaction: categorical (qualitative, ordinal). (v) Teaching standards: categorical (qualitative). (vi) Temperature measured in Fahrenheit: continuous (quantitative).

1.6 Explain in detail the main aspects of a Statistical problem?

The main aspects of a statistical problem are: (i) a clear definition of the problem

and its objectives; (ii) the design of the experiment or survey; (iii) the collection of data; (iv) the organisation and presentation of the data; (v) the analysis of the data; and (vi) the interpretation of the results and the decisions based on them. Each step must be settled before the next begins, or the conclusions will be unreliable.

1.7 Define Descriptive and Inferential Statistics and differentiate between them.

Descriptive Statistics consists of the methods of organising, summarising and presenting data in tables, graphs and averages (mean, median, mode) so that the main features of the data are seen at a glance; it describes only the data actually collected. Inferential Statistics uses the sample to draw conclusions (inferences) about the population, with the help of probability theory, e.g. estimating a population mean from a sample mean. Difference: descriptive statistics stays within the observed data, while inferential statistics goes beyond it to make general statements about the population, with a stated degree of confidence.

1.8 Distinguish between primary and secondary data and give different sources from which these are obtained.

Primary data are collected first-hand by the investigator himself for his own purpose, e.g. by direct personal observation, by interview, by questionnaire sent by mail, by telephone, or through the internet. Secondary data are data already collected by some other person or agency and used by the investigator, e.g. from government organisations (Federal and Provincial Bureau of Statistics, Census organisations), semi-government organisations (municipal committees, banks), teaching and research organisations, research journals and newspapers, and the internet. Primary data are original and usually more accurate for the purpose, but costly; secondary data are cheap and quick but may not fit the purpose exactly.

Exercise 1 - 1.9 Fill in the Blanks

1.9 (i) The purpose of the sample is to draw inference about population .

1.9 (ii) Proportion is always less than or equal to one.

1.9 (iii) The quantity computed from population is called parameter .

1.9 (iv) The quantity computed from sample is called statistic .

1.9 (v) The quantity which does not vary from individual to individual is called constant .

1.9 (vi) Sum of the random errors equal to zero .

1.9 (vii) A variable is called random when its value cannot be exactly determined.

1.9 (viii) A variable that takes numerical values is called quantitative variable.

1.9 (ix) The procedure of inferring about the population characteristics using the sample is called inferential statistics .

1.9 (x) First hand collected data is called primary data .

Exercise 1 - 1.10 True or False

1.10 (i) The value calculated from the population is called parameter.

True

1.10 (ii) Statistics deals with single fact.

False

1.10 (iii) Statistics can be placed in relation to each other.

True

1.10 (iv) Primary data and ungrouped data are same.

False

1.10 (v) A collection of observations is called data.

True

1.10 (vi) A variable can assume the same value.

True

1.10 (vii) A discrete variable can assume finite values between two given limits.

False

1.10 (viii) Height measurements of students is quantitative variable.

True

1.10 (ix) The word Statistics is at present used in four ways.

True

1.10 (x) A small part of the population is called sample.

True

Exercise 1 - Note on the Answers key

Note - discrepancy between the answers above and the book's printed key for 1.10 (ii), (iii), (v) and (viii).

Note on the book's printed Answers key (p. 250): the key marks 1.10 (ii) T, (iii) F, (v) F and (viii) F. Those four key entries conflict with the chapter's own definitions (statistics deals with aggregates, not a single isolated fact; a collection of observations is data; statistics are comparable with each other; height is a quantitative variable) and are treated here as misprints in the key. The answers above follow the definitions. All other 1.10 key entries agree with the answers above, and the 1.9 fill-in answers above agree with the key exactly.

Chapter 2: Representation of Data

Exercise 2 - Questions 2.1 to 2.16

2.1 What are the different methods of representing data?

Data can be represented in three main ways: (i) by tables, in which data are arranged in rows and columns according to a scheme of classification; (ii) by diagrams, such as bar diagrams, pie diagrams and pictograms, which show the data pictorially; (iii) by graphs, such as histograms, frequency polygons, frequency

curves and ogives, which show the data on a coordinate system.

2.2 Define a histogram, a frequency polygon and a frequency curve.

Histogram: a graph of a frequency distribution in which rectangles are drawn on class intervals as bases, with heights proportional to the class frequencies, and with no gaps between consecutive rectangles.

Frequency polygon: a graph obtained by joining the mid-points of the tops of the rectangles of a histogram by straight lines, extended to the horizontal axis at both ends.

Frequency curve: a smooth curve drawn through the points of a frequency polygon, used when the number of observations is large and the class intervals are narrow.

2.3 Define classification and tabulation, and explain the importance of tabulation.

Classification is the arrangement of data into groups or classes according to some common characteristics. Tabulation is the systematic presentation of classified data in rows and columns.

Tabulation is important because it organises data in a compact form, makes comparison easy, brings out the main features of the data, and makes further statistical analysis possible.

2.4 Distinguish between one-way and two-way tables. Also define: (i) classification according to attributes, (ii) class limits, (iii) length of class interval, (iv) class frequency.

A one-way table classifies data according to a single characteristic, for example the number of students in each class of a school. A two-way table classifies data according to two characteristics at the same time, for example students classified by class and by sex.

(i) Classification according to attributes means grouping data by a quality that cannot be measured in numbers, such as sex, religion or nationality. (ii) Class limits are the smallest and largest values that can belong to a class. (iii) The length of a

class interval is the difference between the upper and lower class boundaries. (iv) Class frequency is the number of observations falling in a class.

2.5 The monthly budgets of two families are: Family A: Food Rs. 600, Clothing Rs. 100, House rent Rs. 400, Fuel and lighting Rs. 100, Miscellaneous Rs. 300, Total Rs. 1500. Family B: Food Rs. 800, Clothing Rs. 100, House rent Rs. 500, Fuel and lighting Rs. 100, Miscellaneous Rs. 500, Total Rs. 2000. Represent the data through a suitable diagram.

Step 1: A suitable diagram is a multiple bar diagram, with one pair of bars for each item of expenditure, one bar for Family A and one for Family B.

Step 2: As percentages of the family total, Family A spends: Food 40 percent, Clothing 6.67 percent, House rent 26.67 percent, Fuel and lighting 6.67 percent, Miscellaneous 20 percent.

Step 3: Family B spends: Food 40 percent, Clothing 5 percent, House rent 25 percent, Fuel and lighting 5 percent, Miscellaneous 25 percent.

Step 4: Draw, for each item, two bars side by side with heights proportional to the amounts (or to the percentages), labelled A and B, with a common scale and a title.

Answer: Multiple bar diagram with the percentages above; both families spend the largest share on food (40 percent each), and Family B spends a larger share on house rent and miscellaneous items than Family A.

2.6 Draw a pie diagram for the following expenditure: Food Rs. 4000, Clothing Rs. 1000, House Rent Rs. 2500, Education Rs. 1000, Fuel and light Rs. 600, Miscellaneous Rs. 2000.

Step 1: Total expenditure = 4000 + 1000 + 2500 + 1000 + 600 + 2000 = Rs. 11100.

Step 2: Angle for each item = (item divided by 11100) times 360 degrees. Food: 129.73 degrees; Clothing: 32.43 degrees; House Rent: 81.08 degrees; Education: 32.43 degrees; Fuel and light: 19.46 degrees; Miscellaneous: 64.86 degrees.

Step 3: Draw a circle and divide it into sectors with these central angles, in order, labelling each sector with the item and its angle.

Answer: Pie diagram with sector angles 129.73, 32.43, 81.08, 32.43, 19.46 and 64.86 degrees, adding to 360 degrees.

2.7 Draw a histogram for the following distribution, giving the steps involved. Mid values (X): 32, 37, 42, 47, 52, 57, 62, 67. Frequencies (f): 3, 17, 28, 47, 54, 31, 14, 4.

Step 1: The mid values differ by 5, so the class width is 5 and the class boundaries are 29.5-34.5, 34.5-39.5, 39.5-44.5, 44.5-49.5, 49.5-54.5, 54.5-59.5, 59.5-64.5, 64.5-69.5.

Step 2: Draw rectangles on these class intervals as bases, with heights equal to the frequencies 3, 17, 28, 47, 54, 31, 14, 4, with no gaps between consecutive rectangles.

Step 3: Mark the mid values on the horizontal axis, choose a suitable scale on the vertical axis, and give the histogram a title.

Answer: Histogram with the eight classes and frequencies above; the tallest rectangle is the class 49.5-54.5 with frequency 54.

2.8 (i) State the important points to be kept in mind while drawing graphs. The lengths of a sample of 100 leaves are grouped as: mid values 2.2, 2.7, 3.2, 3.7, 4.2, 4.7, 5.2, 5.7, 6.2 with frequencies 3, 5, 8, 12, 18, 24, 20, 8, 2. (ii) (a) Display the data as a frequency polygon. (b) Find the boundaries of the interval whose mid-point is 3.7 cm.

(i) Important points for drawing graphs: give a clear title; choose suitable scales for both axes and mark them; label both axes; plot the points correctly; join the points as the type of graph requires; keep the graph neat and proportional.

The leaf-length data (100 leaves) are: mid values 2.2, 2.7, 3.2, 3.7, 4.2, 4.7, 5.2, 5.7, 6.2 with frequencies 3, 5, 8, 12, 18, 24, 20, 8, 2.

(ii)(a) Frequency polygon: plot the points (mid value, frequency): (2.2, 3), (2.7, 5), (3.2, 8), (3.7, 12), (4.2, 18), (4.7, 24), (5.2, 20), (5.7, 8), (6.2, 2). Join consecutive points by straight lines, and extend the polygon to the horizontal axis at 1.7 and 6.7, one class width beyond the first and last mid values.

(ii)(b) The class interval whose mid-point is 3.7 has boundaries 3.45-3.95, and the printed key gives its class limits as 3.455-3.945, following the convention of the source table. Note: the question says '3.7 cm' while the leaf-length table is in millimetres; the values are given here in the table's units.

2.9 Draw a pie diagram for the following area under crops, total area 500 acres: sugarcane 250 acres, maize 125 acres, wheat 60 acres, other crops 65 acres.

Step 1: Angles: sugarcane = (250/500) times 360 = 180 degrees; maize = (125/500) times 360 = 90 degrees; wheat = (60/500) times 360 = 43.2 degrees; other crops = (65/500) times 360 = 46.8 degrees.

Step 2: Draw a circle divided into sectors of 180, 90, 43.2 and 46.8 degrees, labelled with the crop names.

Answer: Pie diagram with sector angles 180, 90, 43.2 and 46.8 degrees.

2.10 A biologist measures the lengths (in mm) of male and female green lynx spiders. Female lengths (22 values): 5.7, 5.2, 4.7, 5.8, 6.1, 6.3, 7.0, 5.7, 7.5, 6.4, 6.5, 4.7, 6.2, 5.4, 6.2, 5.8, 4.8, 5.9, 5.2, 6.8, 5.6, 5.5. Male lengths (20 values): 8.2, 9.9, 5.9, 8.4, 9.6, 7.0, 6.6, 7.8, 7.5, 9.8, 6.3, 8.3, 8.0, 9.1, 6.3, 8.4, 8.7, 7.4, 7.9, 10.2. Advise the biologist on how to compare the two sets.

Step 1: Female mean = 129.0 divided by 22 = 5.86 mm. Male mean = 161.3 divided by 20 = 8.07 mm. On average, males are longer by about 2.20 mm.

Step 2: Female lengths range from 4.7 mm to 7.5 mm; male lengths range from 5.9 mm to 10.2 mm. The ranges overlap, but the male range lies higher.

Step 3: Advice: form a grouped frequency distribution for each sex with the same class intervals, and draw the two histograms (or frequency polygons) on the same axes. The comparison of the means, ranges and the shapes of the distributions then shows clearly that males are systematically longer than females.

Answer: Compare equal-interval frequency distributions and their graphs; males average 8.07 mm against 5.86 mm for females.

2.11 (i) Define tabulation and state its main steps. (ii) What is a frequency distribution, and how is it constructed?

(i) Tabulation is the systematic arrangement of classified data in rows and columns.

Its main steps are: decide the title and the basis of classification; choose the rows (stubs) and columns (captions); decide the number of classes and their limits; enter the data with totals and sub-totals; and add footnotes and the source where needed.

(ii) A frequency distribution is a table showing the values (or classes) of a variable together with the number of observations in each value or class. To construct one: find the range of the data; decide the number of classes and the class width; fix the class limits and boundaries; tally the observations into the classes; and record the frequency of each class, with the total.

2.12 The lifetimes (in minutes, to the nearest tenth) of 50 sprayed insects are: 1.2, 2.2, 0.7, 3.9, 1.7, 1.9, 1.4, 1.8, 2.0, 4.3, 2.5, 0.9, 3.4, 2.8, 3.7, 3.5, 0.4, 2.8, 1.1, 0.2, 3.9, 6.3, 2.5, 2.1, 1.3, 2.1, 0.3, 0.4, 2.4, 2.1, 3.5, 2.9, 1.2, 5.3, 1.7, 2.7, 1.8, 4.8, 3.2, 1.6, 2.6, 1.8, 2.3, 1.3, 3.1, 1.5, 2.6, 5.9, 2.0, 2.3. Using 8 intervals with the lowest starting at 0.1: (i) form the frequency and cumulative frequency distributions, (ii) draw the histogram and the frequency polygon.

Step 1: (i) With width 0.8, the classes and frequencies are: 0.1-0.8: 5; 0.9-1.6: 9; 1.7-2.4: 15; 2.5-3.2: 10; 3.3-4.0: 6; 4.1-4.8: 2; 4.9-5.6: 1; 5.7-6.4: 2. The total is 50.

Step 2: Cumulative frequencies (less than type) are: 5, 14, 29, 39, 45, 47, 48, 50.

Step 3: (ii) Histogram: use class boundaries 0.05-0.85, 0.85-1.65, 1.65-2.45, 2.45-3.25, 3.25-4.05, 4.05-4.85, 4.85-5.65, 5.65-6.45, with rectangle heights equal to the frequencies above.

Step 4: Frequency polygon: join the mid-points of the tops of the rectangles, that is the points (0.45, 5), (1.25, 9), (2.05, 15), (2.85, 10), (3.65, 6), (4.45, 2), (5.25, 1), (6.05, 2), by straight lines, extended to the axis at -0.35 and 6.85.

Answer: Frequency table and cumulative frequencies as above, with the histogram and polygon drawn from them.

2.13 (i) What are the advantages of diagrammatic representation of data? (ii) Define: (a) bar diagram, (b) subdivided bar diagram, (c) multiple bar diagram.

(i) Diagrams make data easy to understand at a glance; they attract attention; they make comparison easy; they save time; they reveal patterns and trends that are hard to see in a table; and they are useful for presenting data to non-experts.

(ii) (a) A bar diagram represents data by bars of equal width whose lengths are proportional to the values. (b) A subdivided bar diagram uses a single bar for each category, divided into segments proportional to the components, so that both the total and its parts are shown. (c) A multiple bar diagram places two or more bars side by side for each category, so that two or more related series can be compared.

2.14 Draw a bar diagram for the following data. Year: 1950, 1951, 1952, 1953, 1954, 1955, 1956, 1957. Value in Rs. (000): 1010, 2050, 3458, 1980, 2300, 1295, 1520, 1070.

Step 1: Draw one bar for each year, all of equal width, with heights proportional to the values 1010, 2050, 3458, 1980, 2300, 1295, 1520 and 1070 on a common scale.

Step 2: Mark the years on the horizontal axis and the values on the vertical axis, and give the diagram a title.

Answer: Bar diagram with the tallest bar for 1952 (Rs. 3458 thousand) and the shortest for 1950 (Rs. 1010 thousand).

2.15 Draw a sub-divided bar diagram for the male and female population of four

divisions of Punjab in 1961 (in thousands): Lahore Male 35, Female 30, Both 65; Multan Male 35, Female 31, Both 66; Sargodha Male 32, Female 28, Both 60; Rawalpindi Male 21, Female 19, Both 40.

Step 1: Draw one bar for each division, with total heights 65, 66, 60 and 40 on a common scale.

Step 2: Divide each bar into two segments: Lahore 35 male and 30 female; Multan 35 male and 31 female; Sargodha 32 male and 28 female; Rawalpindi 21 male and 19 female. Shade the two segments differently and add a key.

Answer: Sub-divided bar diagram with the four bars and segments above; in every division the male segment is larger than the female segment.

2.16 Draw a scatter plot and a bivariate frequency distribution for the following (age, weight) pairs: (20, 60), (15, 55), (14, 45), (17, 60), (16, 48), (22, 70), (16, 63), (14, 55), (18, 57), (19, 67), (21, 67), (17, 65), (13, 60), (15, 60), (17, 49), (19, 65), (23, 73), (21, 65), (22, 70), (14, 50), (16, 60), (19, 59), (15, 62), (17, 59), (24, 75).

Step 1: Scatter plot: plot age on the horizontal axis and weight on the vertical axis, marking one point for each of the 25 pairs above. The points rise from left to right, showing that weight tends to increase with age.

Step 2: Bivariate frequency distribution, using age classes 13-15, 16-18, 19-21, 22-24 and weight classes 45-54, 55-64, 65-75. Counting the pairs: age 13-15 gives 2, 5, 0; age 16-18 gives 2, 5, 1; age 19-21 gives 0, 2, 4; age 22-24 gives 0, 0, 4.

Step 3: The table, with rows for age and columns for weight, is: 13-15: 2, 5, 0 (total 7); 16-18: 2, 5, 1 (total 8); 19-21: 0, 2, 4 (total 6); 22-24: 0, 0, 4 (total 4). Column

totals are 4, 12, 9, and the grand total is 25.

Answer: Scatter plot of the 25 points as above, and the bivariate table above. The class intervals are a solver's choice; any reasonable grouping that counts the same 25 pairs is correct.

Exercise 2 - 2.17 Fill in the Blanks

2.17 (i) Classification is the process of arranging data according to some common characteristics.

2.17 (ii) A table has at least four parts.

2.17 (iii) In an open-end frequency distribution, either the _____ class limit of _____ group or the upper limit of the _____ class is not given.

In an open-end frequency distribution, either the lower class limit of the first group or the upper limit of the last class is not given.

2.17 (iv) In a histogram, with unequal class intervals, the area of each rectangle is proportional to the class frequency.

2.17 (v) An ogive is a cumulative frequency polygon.

2.17 (vi) A frequency table can be represented graphically by a histogram .

2.17 (vii) A histogram is a _____ bar chart with _____ space between its bars.

A histogram is a vertical bar chart with no space between its bars.

2.17 (viii) The area of each bar is proportional to the frequency it represents.

2.17 (ix) If the mid-points of the tops of the consecutive bars in a histogram are joined by straight lines, a _____ is obtained.

If the mid-points of the tops of the consecutive bars in a histogram are joined by straight lines, a frequency polygon is obtained.

Exercise 2 - 2.18 True or False

2.18 (i) Grouped data and primary data are the same.

False

2.18 (ii) The class mark is also named as the mid-point.

True

2.18 (iii) A table has at least three parts.

False

2.18 (iv) The graph of a time series is called a histogram.

False

2.18 (v) Cumulative frequencies are decreasing.

False

2.18 (vi) The data 10, 5, 7, 6, 4 is an example of grouped data.

False

2.18 (vii) The two-fold division is also named as dichotomy.

True

2.18 (viii) Data can be presented by means of a graph.

True

2.18 (ix) A graph of a cumulative frequency curve is called a polygon.

False

2.18 (x) In constructing a histogram, mid-points are to be taken along the x-axis.

False

Exercise 2 - Note on the Answers key

Note on the book's printed Answers key (p. 250): the key marks 2.18 (iv) True.

The statement in 2.18 (iv) says that the graph of a time series is called a histogram. The standard name for the graph of a time series is a historigram, while a histogram is the graph of a frequency distribution. The answer given here is therefore False,

and the key's True is treated as a misprint, following the same policy as the note in Exercise 1.

Chapter 3: Measure of Location

Exercise 3 - Questions 3.1 to 3.30

3.1 (i) Define A.M., G.M. and H.M., and state the situation in which each is used. (ii) State the relation between A.M., G.M. and H.M., and the situation in which they are equal.

(i) Arithmetic mean (A.M.): the sum of all observations divided by their number. It is used when all values carry equal importance and the total matters, as for average marks or average income.

Geometric mean (G.M.): the nth root of the product of n observations. It is used for averaging ratios, percentages and rates of change, such as population growth or compound interest.

Harmonic mean (H.M.): the reciprocal of the arithmetic mean of the reciprocals. It is used for averaging rates, such as speed over equal distances or price per unit.

(ii) For any set of positive observations, A.M. >= G.M. >= H.M. The three are equal when all the observations are equal.

3.2 Find the geometric mean of 50, 67, 39, 40, 36, 60, 54, 43.

Step 1: G.M. = (50 times 67 times 39 times 40 times 36 times 60 times 54 times 43) to the power 1/8.

Step 2: Taking logarithms (base 10): log values are 1.6990, 1.8261, 1.5911, 1.6021, 1.5563, 1.7782, 1.7324, 1.6335; their sum is 13.4187 and their mean is 1.6773.

Step 3: Antilog of 1.6773 = 47.5675.

Answer: 47.5675.

3.3 A man travels 100 km in 5 stages at equal intervals, with speeds 10, 16, 20, 14 and 15 km per hour. Find his average speed.

Step 1: The stages cover equal distances, so the average speed is the harmonic mean.

Step 2: H.M. = 5 divided by (1/10 + 1/16 + 1/20 + 1/14 + 1/15) = 5 divided by 0.35059 = 14.2615. Note: the printed key gives 14.2614, a last-digit rounding difference.

Answer: 14.2615 km per hour.

3.4 Calculate the mean, median and mode for Table 2.5 (Chapter 2): the number of rotten potatoes in 20 samples. Values (X): 0, 1, 2, 3, 4 with frequencies (f): 5, 6, 4, 4, 1.

Step 1: Mean = (0 times 5 + 1 times 6 + 2 times 4 + 3 times 4 + 4 times 1) divided by 20 = 30/20 = 1.5.

Step 2: Cumulative frequencies are 5, 11, 15, 19, 20. The median position is (20 + 1)/2 = 10.5th observation, which lies in the cumulative frequency 11, so the median is 1.

Step 3: The highest frequency is 6, for the value 1, so the mode is 1.

Answer: Mean 1.5, Median 1, Mode 1.

3.5 Calculate the mean, median and mode for the grouped data of Table 2.7 (Chapter 2): the ages of 51 members of a locality. Classes: 2-4, 4-8, 8-12, 12-16, 16-22, 22-30 with frequencies 5, 10, 12, 14, 6, 4.

Step 1: Mid values are 3, 6, 10, 14, 19, 26. Sum of f times mid value = 5 times 3 + 10 times 6 + 12 times 10 + 14 times 14 + 6 times 19 + 4 times 26 = 15 + 60 + 120 + 196 + 114 + 104 = 609. Mean = 609/51 = 11.94.

Step 2: Cumulative frequencies are 5, 15, 27, 41, 47, 51. The median position is 51/2 = 25.5, which lies in the class 8-12. Median = 8 + ((25.5 - 15)/12) times 4 = 8 + 3.5 = 11.5.

Step 3: The modal class is 12-16 (frequency 14). Mode = 12 + ((14 - 12) divided by (2 times 14 - 12 - 6)) times 4 = 12 + 0.8 = 12.8.

Step 4: Note: the printed key gives the mean as 11.67, but the frequencies and mid values above give 609/51 = 11.94. The median (11.5) and mode (12.8) agree with the key, so the key's mean is treated as a misprint.

Answer: Mean 11.94, Median 11.5, Mode 12.8.

3.6 For the data of exercise 2.8 (b) (leaf lengths of 100 leaves): mid values 2.2, 2.7, 3.2, 3.7, 4.2, 4.7, 5.2, 5.7, 6.2 with frequencies 3, 5, 8, 12, 18, 24, 20, 8, 2: (i) estimate the mean leaf length, (ii) construct the cumulative frequency table and estimate the sample median.

Step 1: (i) Sum of f times mid value = 2.2 times 3 + 2.7 times 5 + 3.2 times 8 + 3.7

times 12 + 4.2 times 18 + 4.7 times 24 + 5.2 times 20 + 5.7 times 8 + 6.2 times 2 = 6.6 + 13.5 + 25.6 + 44.4 + 75.6 + 112.8 + 104 + 45.6 + 12.4 = 440.5. Mean = 440.5/100 = 4.405.

Step 2: (ii) Cumulative frequencies are 3, 8, 16, 28, 46, 70, 90, 98, 100. The median position is 100/2 = 50, which lies in the class with boundaries 4.45-4.95 (frequency 24, preceding cumulative frequency 46).

Step 3: Median = 4.45 + ((50 - 46)/24) times 0.5 = 4.45 + 0.083 = 4.533.

Answer: (i) Mean 4.405, (ii) Median 4.533.

3.7 Define the weighted mean. When is it preferred to the ordinary mean, and why?

The weighted mean gives each observation a weight according to its importance, and is the sum of weight times observation divided by the sum of the weights.

It is preferred to the ordinary mean when observations are not equally important, for example when combining marks in subjects with different credit hours, or prices of commodities with different consumption levels. The ordinary mean would give a minor item the same influence as a major one.

3.8 Define the mode of a frequency distribution and compare it with the other averages.

The mode of a frequency distribution is the value (or class) with the highest frequency.

Compared with the mean and median, the mode is the easiest to understand and is not affected by extreme values, but it may not exist or may not be unique, and it does not use all the observations. It is the most suitable average for qualitative data and for the most common size, such as the most popular shoe size.

3.9 (i) State the empirical relation between mean, median and mode for a unimodal, moderately asymmetrical distribution, and the relative positions of the three averages for right-skewed and left-skewed curves.

(i) For a unimodal, moderately asymmetrical distribution, the empirical relation is: Mean - Mode = 3 times (Mean - Median), or Mode = 3 times Median - 2 times Mean. For a right-skewed (positively skewed) distribution, the order is Mode < Median < Mean. For a left-skewed (negatively skewed) distribution, the order is Mean < Median < Mode.

3.9 (ii) If the mean is 45 and the median is 36, find the mode approximately.

Step 1: Mode = 3 times Median - 2 times Mean = 3 times 36 - 2 times 45 = 108 - 90 = 18.

Answer: 18.

3.10 Bilal's salary rises by 10 percent at the end of the first year, by 20 percent in the second year and by 25 percent in the third year, each rise being on the salary at the beginning of that year. Find the equivalent annual percentage increase.

Step 1: The growth factors are 1.10, 1.20 and 1.25. Their geometric mean = (1.10 times 1.20 times 1.25) to the power 1/3 = (1.65) to the power 1/3 = 1.182.

Step 2: Equivalent annual increase = 1.182 - 1 = 0.182, that is 18.2 percent.

Answer: 18.2 percent per year.

3.11 Calculate the mean for the following distribution. Classes: 0-10, 10-40, 40-90, 90-100, 100-105, 105-120, 120-140 with frequencies 40, 110, 150, 200, 120, 30, 20.

Step 1: Mid values are 5, 25, 65, 95, 102.5, 112.5, 130. Total frequency N = 40 +

110 + 150 + 200 + 120 + 30 + 20 = 670.

Step 2: Sum of f times mid value = 40 times 5 + 110 times 25 + 150 times 65 + 200 times 95 + 120 times 102.5 + 30 times 112.5 + 20 times 130 = 200 + 2750 + 9750 + 19000 + 12300 + 3375 + 2600 = 49975.

Step 3: Mean = 49975/670 = 74.59.

Step 4: Note: the printed key gives 74.7024, which would need a total of about 50051 instead of 49975. No clean change in the printed frequencies produces that total, so the key's value is treated as a misprint and the computed 74.59 is given.

Answer: 74.59.

3.12 The following distribution is derived from a working origin. If D = X - 18, find the A.M. and G.M. D: -12, -8, -4, 0, 4, 8, 12, 16 with frequencies 2, 5, 8, 18, 22, 13, 8, 4.

Step 1: Total frequency N = 2 + 5 + 8 + 18 + 22 + 13 + 8 + 4 = 80.

Step 2: Sum of f times D = 2 times (-12) + 5 times (-8) + 8 times (-4) + 18 times 0 + 22 times 4 + 13 times 8 + 8 times 12 + 4 times 16 = -24 - 40 - 32 + 0 + 88 + 104 + 96 + 64 = 256. Mean of D = 256/80 = 3.2, so A.M. = 18 + 3.2 = 21.2.

Step 3: The X values are D + 18: 6, 10, 14, 18, 22, 26, 30, 34 with the same frequencies. Their geometric mean = 20.06 (from the mean of the logarithms of X, weighted by the frequencies).

Answer: A.M. 21.2, G.M. 20.06.

3.13 The reciprocals of 11 values of x are: 0.0500, 0.0454, 0.0400, 0.0333, 0.0285, 0.0232, 0.0213, 0.02000, 0.0182, 0.0151, 0.0143. Find the A.M. and H.M. of x.

Step 1: H.M. = 11 divided by the sum of the reciprocals = 11 divided by 0.3093 = 35.56.

Step 2: The x values are the reciprocals of the given numbers: 20, 22.03, 25, 30.03, 35.09, 43.10, 46.95, 50, 54.95, 66.23, 69.93. Their sum is 463.30, so A.M. = 463.31/11 = 42.12.

Answer: A.M. 42.12, H.M. 35.56.

3.14 The reciprocals of 7 values of x are: 0.0267, 0.0235, 0.0211, 0.0191, 0.0174, 0.0160, 0.0148. Find the H.M. of x.

Step 1: Sum of the reciprocals = 0.0267 + 0.0235 + 0.0211 + 0.0191 + 0.0174 + 0.0160 + 0.0148 = 0.1386.

Step 2: H.M. = 7 divided by 0.1386 = 50.51.

Answer: 50.51.

3.15 Three cities A, B and C are equidistant. Fatima travels from A to B by car at 30 miles per hour and from B to C at 50 miles per hour. Find her average speed.

Step 1: Equal distances are covered at different speeds, so the average speed is the harmonic mean.

Step 2: H.M. = 2 divided by (1/30 + 1/50) = 2 divided by 0.05333 = 37.5.

Answer: 37.5 miles per hour.

3.16 The H.M. and G.M. of two numbers are 3.2 and 4. Find the A.M. and the numbers.

Step 1: For two numbers, A.M. times H.M. = (G.M.) squared, so A.M. = 16/3.2 = 5.

Step 2: Let the numbers be a and b. Then a + b = 2 times 5 = 10 and a times b = (G.M.) squared = 16. Solving, the numbers are 2 and 8.

Answer: A.M. 5; the numbers are 2 and 8.

3.17 The A.M. and G.M. of three numbers are 34 and 18. The G.M. of the first two numbers is 9. Find the numbers.

Step 1: Let the numbers be x1, x2, x3. Then x1 + x2 + x3 = 3 times 34 = 102, and x1 times x2 times x3 = 18 cubed = 5832.

Step 2: The G.M. of the first two is 9, so x1 times x2 = 81. Then x3 = 5832/81 = 72.

Step 3: Also x1 + x2 = 102 - 72 = 30, with x1 times x2 = 81, so x1 and x2 are 3 and 27.

Answer: 3, 27, 72.

3.18 (i) Find the average rate of motion when a body covers the first mile at 10 miles per hour, the next mile at 8 miles per hour and the third mile at 6 miles per

hour. (ii) A population increases by 20 percent in the first decade, by 25 percent in the next decade and by 4 percent in the third decade. Find the average rate of increase.

Step 1: (i) Equal distances at different speeds, so the average rate is the harmonic mean: H.M. = 3 divided by (1/10 + 1/8 + 1/6) = 3 divided by 0.3917 = 7.66 miles per hour.

Step 2: (ii) The growth factors are 1.20, 1.25 and 1.04. Their geometric mean = (1.20 times 1.25 times 1.04) to the power 1/3 = (1.56) to the power 1/3 = 1.1598, so the average rate of increase is 15.98 percent per decade.

Answer: (i) 7.66 miles per hour, (ii) 15.98 percent per decade.

3.19 Calculate the mean, median and mode for the following distribution of maximum loads (in some unit) borne by cables. Classes: 9.3-9.7 (2 cables), 9.8-10.2 (5), 10.3-10.7 (12), 10.8-11.2 (17), 11.3-11.7 (14), 11.8-12.2 (6), 12.3-12.7 (3), 12.8-13.2 (1).

Step 1: Mid values are 9.5, 10.0, 10.5, 11.0, 11.5, 12.0, 12.5, 13.0 with frequencies 2, 5, 12, 17, 14, 6, 3, 1. Total N = 60. Sum of f times mid value = 19 + 50 + 126 + 187 + 161 + 72 + 37.5 + 13 = 665.5. Mean = 665.5/60 = 11.09.

Step 2: Cumulative frequencies are 2, 7, 19, 36, 50, 56, 59, 60. The median position is 30, in the class with boundaries 10.75-11.25 (frequency 17, preceding cumulative frequency 19). Median = 10.75 + ((30 - 19)/17) times 0.5 = 10.75 + 0.32 = 11.07.

Step 3: The modal class is 10.75-11.25 (frequency 17). Mode = 10.75 + ((17 - 12) divided by (2 times 17 - 12 - 14)) times 0.5 = 10.75 + 0.31 = 11.06.

Answer: Mean 11.09, Median 11.07, Mode 11.06.

3.20 Calculate the mean, median, mode, 6th decile and 74th percentile for the following distribution. Classes: 0.7312-0.7313 (10), 0.7314-0.7315 (15), 0.7316-0.7317 (20), 0.7318-0.7319 (25), 0.7320-0.7321 (30), 0.7322-0.7323 (8), 0.7324-0.7325 (2).

Step 1: Mid values are 0.73125, 0.73145, 0.73165, 0.73185, 0.73205, 0.73225, 0.73245 with frequencies 10, 15, 20, 25, 30, 8, 2. Total N = 110. Sum of f times mid value = 80.4979. Mean = 80.4943/110 = 0.7318.

Step 2: Cumulative frequencies are 10, 25, 45, 70, 100, 108, 110. The median position is 55, in the class with boundaries 0.73175-0.73195 (frequency 25, preceding cumulative frequency 45). Median = 0.73175 + ((55 - 45)/25) times 0.0002 = 0.7318.

Step 3: The modal class is 0.73195-0.73215 (frequency 30). Mode = 0.73195 + ((30 - 25) divided by (2 times 30 - 25 - 8)) times 0.0002 = 0.73195 + 0.00004 = 0.7320.

Step 4: 6th decile: position 0.6 times 110 = 66, in the class 0.73175-0.73195. D6 = 0.73175 + ((66 - 45)/25) times 0.0002 = 0.73192.

Step 5: 74th percentile: position 0.74 times 110 = 81.4, in the class 0.73195-0.73215. P74 = 0.73195 + ((81.4 - 70)/30) times 0.0002 = 0.7320.

Step 6: Note: the printed key gives the mean as 0.7317, the mode as 0.7319 and P74 as 0.7319, last-digit differences from the computed 0.7318, 0.7320 and 0.7320. The values above follow the standard grouped-data formulae.

Answer: Mean 0.7318, Median 0.7318, Mode 0.7320, D6 0.73192, P74 0.7320.

3.21 Calculate Q3, D5, P5 and the mode for the following distribution. Groups: 0-4.9 (3), 5-9.9 (4), 10-14.9 (9), 15-19.9 (11), 20-24.9 (15), 25-29.9 (13), 30-34.9 (13), 35-39.9 (5), 40-44.9 (2), 45-49.9 (2).

Step 1: Class boundaries are -0.05-4.95, 4.95-9.95, 9.95-14.95, 14.95-19.95, 19.95-24.95, 24.95-29.95, 29.95-34.95, 34.95-39.95, 39.95-44.95, 44.95-49.95. Total N = 77. Cumulative frequencies are 3, 7, 16, 27, 42, 55, 68, 73, 75, 77.

Step 2: Q3: position 0.75 times 77 = 57.75, in the class 29.95-34.95 (frequency 13, preceding cumulative frequency 55). Q3 = 29.95 + ((57.75 - 55)/13) times 5 = 31.01.

Step 3: D5 (the median): position 0.5 times 77 = 38.5, in the class 19.95-24.95 (frequency 15, preceding cumulative frequency 27). D5 = 19.95 + ((38.5 - 27)/15) times 5 = 23.783.

Step 4: P5: position 0.05 times 77 = 3.85, in the class 4.95-9.95 (frequency 4, preceding cumulative frequency 3). P5 = 4.95 + ((3.85 - 3)/4) times 5 = 6.0125.

Step 5: The modal class is 19.95-24.95 (frequency 15). Mode = 19.95 + ((15 - 11) divided by (2 times 15 - 11 - 13)) times 5 = 19.95 + 3.33 = 23.28.

Answer: Q3 31.01, D5 23.783, P5 6.0125, Mode 23.28.

3.22 The mean of a distribution is 45 and the median is 30. Find the mode approximately.

Step 1: Mode = 3 times Median - 2 times Mean = 3 times 30 - 2 times 45 = 90 - 90 = 0.

Answer: 0.

3.23 A bus covers 200 miles in ten stages at equal intervals, with speeds 10, 15, 20, 75, 20, 30, 40, 50, 30 and 40 miles per hour. Find the average speed.

Step 1: Equal distances at different speeds, so the average speed is the harmonic mean.

Step 2: H.M. = 10 divided by (1/10 + 1/15 + 1/20 + 1/75 + 1/20 + 1/30 + 1/40 + 1/50 + 1/30 + 1/40) = 10 divided by 0.41667 = 24.00.

Step 3: Note: the printed key gives 24.003, a rounding difference from the exact 24.00.

Answer: 24.00 miles per hour.

3.24 The following data are obtained by the substitution u = (x - 136.5)/6. U: -4, -3, -2, -1, 0, 1, 2, 3 with frequencies 2, 5, 8, 18, 22, 13, 8, 4. Find the H.M. of x.

Step 1: The x values are 136.5 + 6u: 112.5, 118.5, 124.5, 130.5, 136.5, 142.5, 148.5, 154.5. Total N = 80.

Step 2: Sum of f divided by x = 2/112.5 + 5/118.5 + 8/124.5 + 18/130.5 + 22/136.5 + 13/142.5 + 8/148.5 + 4/154.5 = 0.5943.

Step 3: H.M. = 80 divided by 0.5943 = 134.607.

Answer: 134.607.

3.25 Find the weighted mean of the marks with the given weights. English 73 (weight 4), Urdu 82 (weight 3), Mathematics 80 (weight 3), Statistics 57 (weight 2), Physics 62 (weight 2).

Step 1: Sum of weight times marks = 4 times 73 + 3 times 82 + 3 times 80 + 2 times 57 + 2 times 62 = 292 + 246 + 240 + 114 + 124 = 1016.

Step 2: Sum of weights = 4 + 3 + 3 + 2 + 2 = 14. Weighted mean = 1016/14 = 72.57.

Answer: 72.57.

3.26 Find the weighted mean for the following data. Food: expenditure 290, weight 7.5. Rent: 54, weight 2. Clothing: 98, weight 1.5. Fuel and light: 75, weight 1.0. Other items: 75, weight 0.5.

Step 1: Sum of weight times expenditure = 7.5 times 290 + 2 times 54 + 1.5 times 98 + 1.0 times 75 + 0.5 times 75 = 2175 + 108 + 147 + 75 + 37.5 = 2542.5.

Step 2: Sum of weights = 7.5 + 2 + 1.5 + 1.0 + 0.5 = 12.5. Weighted mean = 2542.5/12.5 = 203.4.

Answer: 203.4.

3.27 For a set of values, the sums of deviations from 15, 18 and 211 are 5, 0 and -21 respectively. What is the value of the A.M., and why?

The three sums cannot all hold for one set of values: from the second sum alone,

the mean is 18, because the sum of deviations from 18 is zero. (The first sum would give 3n = 5 and the third would give a different value again, which no whole number n satisfies.)

The mean is therefore 18, following the second sum, which is also the reasoning of the printed key. The other two sums are inconsistent with it and are treated as misprints in the question.

3.28 The A.M. of 15 values is 20. Adding 3 more values, the mean remains 20. Find the three new values if they are in the ratio 3:2:1.

Step 1: The sum of the original 15 values is 15 times 20 = 300. The sum of all 18 values is 18 times 20 = 360, so the sum of the three new values is 60.

Step 2: Dividing 60 in the ratio 3:2:1 gives 30, 20 and 10.

Answer: 30, 20, 10.

3.29 State whether the following are true or false: (i) In a symmetrical distribution, mean, median and mode are equal. (ii) The algebraic sum of deviations from the mean is always zero. (iii) The median is affected by extreme observations. (iv) The mode is not affected by extreme observations. (v) A frequency polygon is an increasing function. (vi) For highly skewed distributions, the median is preferred over the mean. (vii) The mean remains unchanged if a constant is added to each observation.

(i) True: in a symmetrical distribution the mean, median and mode are equal. (ii) True: the algebraic sum of deviations from the mean is always zero. (iii) False: the median is not affected by extreme observations. (iv) True: the mode is not affected by extreme observations. (v) False: a frequency polygon is not an increasing function; it rises and falls with the frequencies. (vi) True: for highly skewed

distributions the median is preferred to the mean. (vii) False: adding a constant to each observation increases the mean by that constant; it does not remain unchanged.

3.30 State the most suitable average for: (i) marks in an examination, (ii) growth rate of population of different cities, (iii) height of students, (iv) size of agricultural holdings, (v) increase in salaries, (vi) IQ level of students in a class.

(i) Marks in an examination: Arithmetic Mean. (ii) Growth rate of population of different cities: Geometric Mean, because growth over time compounds; averaging compound growth rates is a geometric-mean operation, and the book itself uses the geometric mean for population increase in 3.18 (ii). Note: the printed key gives 'Harmonic Mean' for (ii), which is treated as a misprint. (iii) Height of students: Arithmetic Mean. (iv) Size of agricultural holdings: Median. (v) Increase in salaries: Geometric Mean. (vi) IQ level of students in a class: Arithmetic Mean.

Exercise 3 - 3.31 Fill in the Blanks

3.31 (i) The average obtained to represent a data measures of central tendency is called an average.

3.31 (ii) A good average should not be affected by extreme values.

3.31 (iii) The sum of deviations from the mean is always zero .

3.31 (iv) The median divides ordered data into _____ parts.

The median divides ordered data into two equal parts.

3.31 (v) For estimating the average rate of change of population, G.M. is a better average.

3.31 (vi) The mean and median of two values are always equal .

3.31 (vii) In qualitative data, the most suitable average is the mode .

3.31 (viii) A distribution having two modes is called a bimodal distribution.

3.31 (ix) In a symmetrical distribution, the mean, median and mode are identical .

3.31 (x) If extreme large or small values are changed, the values of _____ are not affected.

If extreme large or small values are changed, the values of the median and mode are not affected.

Exercise 3 - 3.32 True or False

3.32 (i) The sum of deviations from the mean is minimum.

False

3.32 (ii) The geometric mean is possible only for negative values.

False

3.32 (iii) The median divides data into two halves.

True

3.32 (iv) The third quartile is the median.

False

3.32 (v) A distribution with only one mode is uni-modal.

True

3.32 (vi) The arithmetic mean depends on all the values.

True

3.32 (vii) In a symmetrical distribution, the mean, median and mode are not equal.

False

3.32 (viii) The harmonic mean can be calculated if any value is zero.

False

3.32 (ix) The median is not affected by extreme values.

True

3.32 (x) The geometric mean cannot be calculated if any value is negative.

True

Chapter 4: Measure of Dispersion

Exercise 4 - Questions 4.1 to 4.38

4.1 Define dispersion. What are the most usual methods of measuring dispersion? Give the advantages and disadvantages of each.

Dispersion is the scatter or variation of the values of a variable about an average value. The most usual measures of dispersion are the range, the quartile deviation, the mean deviation and the standard deviation.

Range is simple and easy to compute but it uses only the two extreme values. Quartile deviation uses the middle half of the data and is not affected by extremes, but it ignores half the observations. Mean deviation uses all observations but its absolute values make it awkward for further algebra. Standard deviation uses all observations and is the most suitable for further mathematical treatment, but it is affected by extreme values.

4.2 Define mean deviation and its coefficient. Give the advantages and uses of mean deviation.

Mean deviation is the mean of the absolute deviations of the values from an average, usually the mean or the median: M.D. = sum of f times |x - average| divided by N. Its coefficient is the mean deviation divided by the average used, and it is a relative measure.

Advantages: it is based on all observations, it is easy to understand, and it is less affected by extreme values than the standard deviation. Uses: it is used in forecasting, in quality control, and wherever the average absolute departure from an average is the quantity of interest.

4.3 (i) Define semi-interquartile range. (ii) Define range and give its uses.

(i) Semi-interquartile range, also called quartile deviation, is half the difference between the third and first quartiles: Q.D. = (Q3 - Q1) divided by 2. It measures the

spread of the middle half of the data.

(ii) Range is the difference between the largest and the smallest value in the data. It is used for a quick, rough idea of spread, for example in quality control charts and in weather reports, but it depends only on the two extreme values.

4.4 Distinguish between absolute and relative measures of dispersion, and state the properties of the standard deviation.

Absolute dispersion is measured in the units of the data itself, for example the standard deviation in kilograms. Relative dispersion is a pure number, free from units, formed by dividing an absolute measure by a suitable average, for example the coefficient of variation. Relative measures are used to compare the variability of two distributions measured in different units or with different averages.

Properties of the standard deviation: it is based on all observations; it is independent of the origin but not of the scale (adding a constant leaves it unchanged, multiplying by a constant multiplies it by the absolute value of the constant); it is the smallest root-mean-square deviation, being smaller than the root-mean-square deviation about any other value; and the variance of a sum of independent variables is the sum of their variances.

4.5 (i) State the various measures of dispersion and their formulae. (ii) For the distribution of marks: 30-39 (8 students), 40-49 (87), 50-59 (190), 60-69 (86), 70-79 (20), calculate (a) the quartile deviation, (b) the coefficient of skewness.

Step 1: (i) The measures are: range = largest value - smallest value; quartile deviation = (Q3 - Q1)/2; mean deviation = sum of f times |x - average| divided by N; standard deviation = square root of (sum of f times (x - mean)^2 divided by N). Their coefficients (relative measures) are obtained by dividing by a suitable average, for example coefficient of quartile deviation = Q.D. divided by ((Q3 + Q1)/2).

Step 2: (ii) Use class boundaries 29.5-39.5, 39.5-49.5, 49.5-59.5, 59.5-69.5, 69.5-79.5, with N = 391.

Step 3: (a) Q1 position = 391/4 = 97.75, in the class 49.5-59.5: Q1 = 49.5 + ((97.75 - 95)/190) times 10 = 49.645. Q3 position = 3 times 391/4 = 293.25, in the class 59.5-69.5: Q3 = 59.5 + ((293.25 - 285)/86) times 10 = 60.459. Quartile deviation = (60.459 - 49.645)/2 = 5.41. (The printed key gives 5.38, a small difference from quartile rounding.)

Step 4: (b) Median = 49.5 + ((195.5 - 95)/190) times 10 = 54.79. Bowley's coefficient of skewness = (Q3 + Q1 - 2 times median) divided by (Q3 - Q1) = (60.459 + 49.645 - 109.58)/10.814 = 0.048, about 0.05.

Answer: (a) Q.D. = 5.41, (b) Coefficient of skewness = 0.05.

4.6 For the distribution of weights (kg): 50-53 (23), 53-56 (24), 56-59 (39), 59-62 (46), 62-65 (54), 65-68 (66), 68-71 (49), 71-74 (38), 74-77 (21), 77-80 (12), find the mean deviation from the mean, the mean coefficient of dispersion, the variance, and also the range, the quartile deviation and the coefficient of quartile deviation.

Step 1: Class mid-points are 51.5, 54.5, 57.5, 60.5, 63.5, 66.5, 69.5, 72.5, 75.5, 78.5, with N = 372. Mean = 64.58.

Step 2: Mean deviation from the mean = sum of f times |x - 64.58| divided by 372 = 5.73.

Step 3: Mean coefficient of dispersion = 5.73 divided by 64.58 = 0.09.

Step 4: Variance = sum of f times (x - mean)^2 divided by 372 = 47.85.

Step 5: Range = 80 - 50 = 30.

Step 6: Q1 = 59.46 (class 59-62) and Q3 = 69.65 (class 68-71), so quartile deviation = (69.65 - 59.46)/2 = 5.10. (The printed key gives 5.30; the value from the quartile formula with these classes is 5.10.) Coefficient of quartile deviation = 5.10 divided by ((69.65 + 59.46)/2) = 0.08.

Answer: M.D. = 5.73, Coefficient of M.D. = 0.09, Variance = 47.85, Range = 30, Q.D. = 5.10, Coefficient of Q.D. = 0.08.

4.7 For the distribution: groups 25-50 (10), 50-75 (12), 75-100 (16), 100-125 (17), 125-150 (20), 150-175 (18), find the quartile deviation and its coefficient.

Step 1: N = 93. Q1 position = 93/4 = 23.25, in the class 75-100: Q1 = 75 + ((23.25 - 22)/16) times 25 = 76.95.

Step 2: Q3 position = 3 times 93/4 = 69.75, in the class 125-150: Q3 = 125 + ((69.75 - 55)/20) times 25 = 143.44.

Step 3: Quartile deviation = (143.44 - 76.95)/2 = 33.24, about 33.25.

Step 4: Coefficient of quartile deviation = 33.24 divided by ((143.44 + 76.95)/2) = 33.24/110.20 = 0.302.

Answer: Q.D. = 33.25, Coefficient of Q.D. = 0.302.

4.8 For the distribution: y = 525, 500, 475, 450, 425, 400, 375 with frequencies 24, 35, 46, 37, 47, 34, 22, find the standard deviation, variance and coefficient of variation.

Step 1: N = 245. Mean = (525 times 24 + 500 times 35 + 475 times 46 + 450 times 37 + 425 times 47 + 400 times 34 + 375 times 22)/245 = 110425/245 = 450.71.

Step 2: Variance = sum of f times (y - mean)^2 divided by 245 = 1996.93.

Step 3: Standard deviation = square root of 1996.93 = 44.687.

Step 4: Coefficient of variation = (44.687/450.71) times 100 = 9.91 percent.

Answer: S.D. = 44.687, Variance = 1996.93, C.V. = 9.91 percent.

4.9 A set of 10 values has mean 25.2 and standard deviation 3.72. Another set of 15 values has mean 25.2 and standard deviation 4.05. Find the combined standard deviation.

Step 1: Since the two means are equal (25.2), the combined mean is also 25.2, and the combined variance is the weighted mean of the two variances.

Step 2: Combined variance = (10 times 3.72^2 + 15 times 4.05^2) divided by 25 = (138.384 + 246.038)/25 = 15.377.

Step 3: Combined standard deviation = square root of 15.377 = 3.92.

Answer: 3.92. Note: the printed key gives 2.49, which cannot be correct, because when the means are equal the combined variance lies between the two variances,

so the combined standard deviation must lie between 3.72 and 4.05.

4.10 The mean height of 50 boys is 59.5 with standard deviation 8.38, and the mean height of 40 girls is 54.0 with standard deviation 8.23. Find the combined mean and combined standard deviation.

Step 1: Combined mean = (50 times 59.5 + 40 times 54.0) divided by 90 = 5135/90 = 57.06.

Step 2: Combined variance = (50 times (8.38^2 + (59.5 - 57.06)^2) + 40 times (8.23^2 + (54.0 - 57.06)^2)) divided by 90 = 76.59.

Step 3: Combined standard deviation = square root of 76.59 = 8.75.

Answer: Combined mean = 57.06, Combined S.D. = 8.75.

4.11 The numbers 3, 6, 1, 7, 2, 5 are multiplied by 2 and then 5 is added to each, giving 11, 17, 7, 19, 9, 15. Find the relationship between the standard deviations and the means of the two sets.

Step 1: Mean of the first set = (3 + 6 + 1 + 7 + 2 + 5)/6 = 24/6 = 4. Mean of the second set = (11 + 17 + 7 + 19 + 9 + 15)/6 = 78/6 = 13 = 2 times 4 + 5.

Step 2: In general, if Y = 2X + 5 then the mean of Y is 2 times the mean of X plus 5, and the standard deviation of Y is 2 times the standard deviation of X, because adding a constant does not change the spread while multiplying by 2 doubles every deviation.

Step 3: Here S.D. of X = square root of 28/6 = 2.16, and S.D. of Y = 4.32 = 2 times

2.16.

Answer: Mean(Y) = 2 Mean(X) + 5 and S.D.(Y) = 2 S.D.(X). Note: the printed key writes S.D.(Y) = S.D.(X), but multiplying every value by 2 must double the standard deviation, so the key's equality is a misprint.

4.12 The scores 70, 50, 60, 70, 50 are changed by (i) adding 10 points to each score, (ii) increasing each score by 10 percent. Find the effect on the mean and the standard deviation in each case.

Step 1: (i) Adding 10 to every score adds 10 to the mean and leaves the standard deviation unchanged, because every deviation from the mean is unchanged. So new mean = old mean + 10, new S.D. = old S.D.

Step 2: (ii) Increasing every score by 10 percent multiplies every score by 110/100, so the new mean is (110/100) times the old mean and the new standard deviation is (110/100) times the old standard deviation.

Answer: (i) Mean increases by 10, S.D. unchanged; (ii) Mean and S.D. are both multiplied by 110/100.

4.13 Find the mean wages and the coefficient of variation for two factories. Wages classes 30-35, 35-40, 40-45, 45-50, 50-55, 55-60; Factory A frequencies 12, 18, 29, 32, 16, 8; Factory B frequencies 4, 10, 31, 67, 35, 15.

Step 1: Class mid-points are 32.5, 37.5, 42.5, 47.5, 52.5, 57.5.

Step 2: Factory A: N = 115, mean = 44.50, standard deviation = 6.85, so C.V. = (6.85/44.50) times 100 = 15.38 percent. (The printed key gives the mean as 44.47,

a small rounding difference.)

Step 3: Factory B: N = 162, mean = 47.56, standard deviation = 5.58, so C.V. = (5.58/47.56) times 100 = 11.74 percent. (The printed key gives 13.32 percent, which does not follow from its own mean and the data; the recomputed value is 11.74 percent.)

Step 4: Factory B has the smaller coefficient of variation, so wages in Factory B are more consistent.

Answer: Factory A: mean 44.50, C.V. 15.38 percent; Factory B: mean 47.56, C.V. 11.74 percent; Factory B is more consistent.

4.14 Find the median and the mean deviation from the median for the distribution of daily wages: wage 6 (5 servants), 8 (10), 10 (18), 12 (20), 14 (22), 16 (14), 18 (7), 20 (3), 22 (1).

Step 1: N = 100. Treating each wage value as a class of width 2 (boundaries 11-13 for wage 12), the median position 50 falls in the wage-12 class: median = 11 + ((50 - 33)/20) times 2 = 12.7.

Step 2: Mean deviation from the median = sum of f times |wage - 12.7| divided by 100 = 286.2/100 = 2.86.

Answer: Median = 12.7, M.D. = 2.86.

4.15 Find the mean deviation about the median for the distribution: groups 100-110 (56), 110-120 (59), 120-130 (61), 130-140 (68), 140-150 (77), 150-160 (59),

160-170 (51), 170-180 (42), 180-190 (36), 190-200 (25).

Step 1: N = 534. The median position 267 falls in the class 140-150: median = 140 + ((267 - 244)/77) times 10 = 142.99.

Step 2: Mean deviation from the median = sum of f times |mid-point - 142.99| divided by 534 = 21.65.

Answer: M.D. (median) = 21.65.

4.16 For the values 3, 5, 7, 13, 15, 17, 23, 27, find the standard deviation by the actual mean method and by the provisional mean method, and compare the results.

Step 1: Sum = 110, so the actual mean = 110/8 = 13.75.

Step 2: By the actual mean: squared deviations are 115.5625, 76.5625, 45.5625, 0.5625, 1.5625, 10.5625, 85.5625, 175.5625, whose sum is 511.5. Variance = 511.5/8 = 63.9375, so S.D. = 8.00 (the key gives 7.99 by rounding).

Step 3: By the provisional mean, say A = 13: deviations are -10, -8, -6, 0, 2, 4, 10, 14, with sum 6 and sum of squares 516. Variance = 516/8 - (6/8)^2 = 64.5 - 0.5625 = 63.9375, the same as before, so S.D. = 8.00.

Answer: S.D. = 8.00 by both methods; the two methods agree, as they must.

4.17 Two makes of television tubes are tested. Tube A has mean life 1496 hours and standard deviation 280 hours; Tube B has mean life 1895 hours and standard deviation 310 hours. Which tube has the greatest (i) absolute dispersion, (ii) relative dispersion?

Step 1: (i) Absolute dispersion is measured by the standard deviation: 310 hours for B is greater than 280 hours for A, so Tube B has the greater absolute dispersion.

Step 2: (ii) Relative dispersion is measured by the coefficient of variation: for A, (280/1496) times 100 = 18.72 percent; for B, (310/1895) times 100 = 16.36 percent. So Tube A has the greater relative dispersion.

Answer: (i) Tube B, (ii) Tube A. Note: the printed key says Tube B for both parts, but 18.72 percent exceeds 16.36 percent, so the relative dispersion is greater for Tube A.

4.18 (i) Define moments about the mean and about an arbitrary value, and state the relation between them. (ii) Define the moment ratios b1 and b2.

(i) The r-th moment about a value a is the mean of (x - a)^r. The moment about the mean is obtained by taking a as the mean; the moment about an arbitrary value a is related to the moments about the mean by shifting: for example, if m1' is the first moment about a, then the mean is a + m1', and the central moments follow from the raw moments by the usual conversion formulae.

(ii) The moment ratios are b1 = (third central moment)^2 divided by (second central moment)^3, which measures skewness, and b2 = (fourth central moment) divided by (second central moment)^2, which measures kurtosis. For a normal distribution b1 = 0 and b2 = 3.

4.19 A computer calculated the mean as 42 and the standard deviation as 5 from 20 observations. Later it was found that two values, copied as 45 and 38, were in fact 35 and 58. Find the correct coefficient of variation.

Step 1: Wrong sum = 20 times 42 = 840. Correct sum = 840 - 45 - 38 + 35 + 58 = 850, so the correct mean = 850/20 = 42.5.

Step 2: Wrong sum of squares = 20 times (25 + 42^2) = 35780. Correct sum of squares = 35780 - 45^2 - 38^2 + 35^2 + 58^2 = 36900.

Step 3: Correct variance = 36900/20 - (42.5)^2 = 1845 - 1806.25 = 38.75, so the correct standard deviation = 6.225.

Step 4: Correct coefficient of variation = (6.225/42.5) times 100 = 14.65 percent (the key gives 14.64 percent, a rounding difference).

Answer: C.V. = 14.65 percent.

4.20 Find the combined coefficient of variation for three groups with frequencies 100, 120, 150, means 5.5, 15.8, 10.5 and standard deviations 2.4, 4.2, 3.7.

Step 1: Combined mean = (100 times 5.5 + 120 times 15.8 + 150 times 10.5) divided by 370 = 4021/370 = 10.87.

Step 2: Combined variance = (100 times (2.4^2 + (5.5 - 10.87)^2) + 120 times (4.2^2 + (15.8 - 10.87)^2) + 150 times (3.7^2 + (10.5 - 10.87)^2)) divided by 370 = 28.57.

Step 3: Combined standard deviation = square root of 28.57 = 5.34, so C.V. = (5.34/10.87) times 100 = 49.17 percent.

Answer: C.V. = 49.17 percent. Note: the printed key gives 44.122 percent, which does not follow from the combined-mean formula with the given groups.

4.21 Find the first four moments about the mean for the marks 45, 32, 37, 46, 39,

36, 41, 48, 36.

Step 1: Sum = 360, so the mean = 360/9 = 40. Deviations from the mean are 5, -8, -3, 6, -1, -4, 1, 8, -4.

Step 2: First moment = 0 (the sum of deviations is 0).

Step 3: Second moment = (25 + 64 + 9 + 36 + 1 + 16 + 1 + 64 + 16)/9 = 232/9 = 25.78 (the key gives 25.7).

Step 4: Third moment = (125 - 512 - 27 + 216 - 1 - 64 + 1 + 512 - 64)/9 = 186/9 = 20.67 (the key gives 20.66).

Step 5: Fourth moment = (625 + 4096 + 81 + 1296 + 1 + 256 + 1 + 4096 + 256)/9 = 10708/9 = 1189.78 (the key gives 1189.7).

Answer: Moments are 0, 25.78, 20.67, 1189.78.

4.22 For the distribution of earnings 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 with frequencies 1, 2, 5, 10, 20, 51, 22, 11, 5, 3, 1, find the first four moments about y = 10 and then the moments about the mean.

Step 1: N = 131. Moments about 10: first = 0.061, second = 2.641, third = 0.565, fourth = 28.382 (the key gives 0.061, 2.64, 0.564, 28.38).

Step 2: Mean = 10 + 0.061 = 10.061.

Step 3: Converting to moments about the mean: first = 0; second = 2.641 - (0.061)^2 = 2.637; third = 0.565 - 3 times 0.061 times 2.641 + 2 times (0.061)^3

= 0.0815 (the key gives 0.0811); fourth = 28.382 - 4 times 0.061 times 0.565 + 6 times (0.061)^2 times 2.641 - 3 times (0.061)^4 = 28.303 (the key gives 28.301).

Answer: Moments about 10: 0.061, 2.641, 0.565, 28.382; moments about the mean: 0, 2.637, 0.0815, 28.303.

4.23 The first three moments about y = 4 are 1, 4 and 10. Find the coefficient of variation and state whether the distribution is symmetrical, positively skewed or negatively skewed.

Step 1: Mean = 4 + 1 = 5.

Step 2: Second central moment = 4 - 1^2 = 3, so standard deviation = square root of 3 = 1.732.

Step 3: Coefficient of variation = (1.732/5) times 100 = 34.64 percent.

Step 4: Third central moment = 10 - 3 times 1 times 4 + 2 times 1^3 = 0, so the distribution is symmetric.

Answer: C.V. = 34.64 percent; the distribution is symmetric.

4.24 The first four moments about y = 17.5 are 0.3, 74, 45 and 12125. Is the distribution leptokurtic or platykurtic?

Step 1: Second central moment = 74 - (0.3)^2 = 73.91.

Step 2: Fourth central moment = 12125 - 4 times 0.3 times 45 + 6 times (0.3)^2 times 74 - 3 times (0.3)^4 = 12110.94.

Step 3: b2 = 12110.94 divided by (73.91)^2 = 2.22 (the key gives 2.158). Since b2 is less than 3, the distribution is platykurtic.

Answer: Platykurtic (b2 = 2.22).

4.25 Find the skewness for: (i) Q2 = 26.01, Q3 = 38.29, Q1 = 13.73; (ii) Mean = 1403 and Mode = 1487.

Step 1: (i) Bowley's coefficient = (Q3 + Q1 - 2 times Q2) divided by (Q3 - Q1) = (38.29 + 13.73 - 52.02)/(38.29 - 13.73) = 0/24.56 = 0, so by the quartiles the distribution is symmetric. (The printed key gives -1.118, which does not follow from these quartiles, since Q3 + Q1 equals 2 times Q2 exactly.)

Step 2: (ii) Mean 1403 is less than mode 1487, so the distribution is negatively skewed.

Answer: (i) 0 (symmetric by quartiles), (ii) negatively skewed.

4.26 Given that sum f = 76, sum fy = 572, sum fy squared = 4848, sum fy cubed = 44240 and sum fy to the fourth = 42580. Find the first four moments about the mean and test the distribution for symmetry and kurtosis.

Step 1: Mean = sum fy divided by sum f = 572/76 = 7.5263.

Step 2: Raw moments about the origin: m1 = 572/76 = 7.5263; m2 = 4848/76 = 63.7895; m3 = 44240/76 = 582.1053; m4 = 42580/76 = 560.2632.

Step 3: Central moments: mu1 = 0; mu2 = m2 - m1 squared = 63.7895 - 56.645 = 7.14; mu3 = m3 - 3 times m1 times m2 + 2 times m1 cubed = 582.1053 - 1440.31

+ 852.67 = -5.53.

Step 4: mu4 from the printed sums = m4 - 4 times m1 times m3 + 6 times m1 squared times m2 - 3 times m1 to the fourth = 560.26 - 17527.0 + 21686.6 - 9629.9 = -4909.99. A fourth moment can never be negative, so the printed sum fy to the fourth = 42580 is certainly misprinted.

Step 5: The most likely misprint is a dropped zero: with sum fy to the fourth = 425800, mu4 = 132.38. Then beta2 = mu4 divided by mu2 squared = 132.38/51.04 = 2.59, which is less than 3, so the distribution is platykurtic.

Step 6: Symmetry: mu3 = -5.53 is negative, so the distribution is negatively skewed. Note: the printed key gives mu2 = 7.139, mu3 = -5.364, mu4 = 125.5705 and labels the distribution positively skewed. That label contradicts the key's own negative mu3, and no clean correction of the printed sums reproduces the key's mu4; the kurtosis conclusion (platykurtic) is the same under the key's mu4 and under the corrected mu4.

Answer: mu1 = 0, mu2 = 7.14, mu3 = -5.53, and the printed fourth sum is inconsistent (it gives an impossible negative mu4); with the dropped zero restored, mu4 = 132.38 and the distribution is negatively skewed and platykurtic.

4.27 Given mean 1403 and mode 1487, is the distribution positively or negatively skewed?

Step 1: The mean is less than the mode (1403 < 1487). For a skewed distribution the mean is pulled towards the longer tail, so a mean below the mode means the longer tail is on the left.

Answer: Negatively skewed.

4.28 If the lower quartile is 142.36, the upper quartile is 167.73 and the median is 153.50, find the coefficient of skewness.

Step 1: Bowley's coefficient = (167.73 + 142.36 - 2 times 153.50) divided by (167.73 - 142.36) = 3.09/25.37 = 0.1218.

Answer: 0.1218 (positively skewed).

4.29 For daily income in rupees, with classes of width Rs. 2 from Rs. 0 to Rs. 18 and frequencies from the lowest class upwards 5, 39, 69, 41, 29, 22, 16, 7, 5, find the coefficient of skewness.

Step 1: N = 233. Classes are 0-2, 2-4, 4-6, 6-8, 8-10, 10-12, 12-14, 14-16, 16-18.

Step 2: Q1 position = 58.25, in class 4-6: Q1 = 4 + ((58.25 - 44)/69) times 2 = 4.413.

Step 3: Median position = 116.5, in class 6-8: median = 6 + ((116.5 - 113)/41) times 2 = 6.171.

Step 4: Q3 position = 174.75, in class 8-10: Q3 = 8 + ((174.75 - 154)/29) times 2 = 9.431.

Step 5: Bowley's coefficient = (9.431 + 4.413 - 2 times 6.171) divided by (9.431 - 4.413) = 1.502/5.018 = 0.299.

Answer: 0.299 (positively skewed).

4.30 The first four moments about x = 2 are 1, 2.5, 5.5 and 16. Find the mean and the coefficient of variation.

Step 1: Mean = 2 + 1 = 3.

Step 2: Second central moment = 2.5 - 1^2 = 1.5, so standard deviation = square root of 1.5 = 1.2247.

Step 3: Coefficient of variation = (1.2247/3) times 100 = 40.82 percent, about 41 percent.

Answer: Mean = 3, C.V. = 41 percent.

4.31 Given that the first four moments about y = 20 are -2, 15, -25 and 80, find the moments about the mean, and b1 and b2.

Step 1: Mean = 20 - 2 = 18.

Step 2: Second central moment = 15 - (-2)^2 = 11.

Step 3: Third central moment = -25 - 3 times (-2) times 15 + 2 times (-2)^3 = -25 + 90 - 16 = 49.

Step 4: Fourth central moment = 80 - 4 times (-2) times (-25) + 6 times (-2)^2 times 15 - 3 times (-2)^4 = 80 - 200 + 360 - 48 = 192.

Step 5: b1 = 49^2 divided by 11^3 = 2401/1331 = 1.80. b2 = 192 divided by 11^2 = 192/121 = 1.59.

Answer: Central moments 0, 11, 49, 192; b1 = 1.80, b2 = 1.59.

4.32 What aspects of a frequency curve do skewness and kurtosis measure?

Skewness measures the lack of symmetry of a frequency curve: whether it has a longer tail on the right (positive skew) or on the left (negative skew). It is measured by coefficients such as Bowley's quartile coefficient or Pearson's (mean - mode) divided by standard deviation, or by the moment ratio b1.

Kurtosis measures the peakedness or flatness of a frequency curve compared with the normal curve. It is measured by b2: b2 = 3 for a normal (mesokurtic) curve, b2 greater than 3 for a peaked (leptokurtic) curve, and b2 less than 3 for a flat-topped (platykurtic) curve.

4.33 The second moments about the mean of two distributions are 9 and 16, and their fourth moments are 230 and 780. Which distribution is (i) leptokurtic, (ii) platykurtic?

Step 1: First distribution: b2 = 230 divided by 9^2 = 230/81 = 2.84, which is less than 3, so it is platykurtic.

Step 2: Second distribution: b2 = 780 divided by 16^2 = 780/256 = 3.05, which is greater than 3, so it is leptokurtic.

Answer: (i) Leptokurtic: the second distribution; (ii) Platykurtic: the first distribution.

4.34 Find the skewness for: (i) median 26.01, quartiles 13.73 and 28.29; (ii) mean 140, mode 148.7; (iii) first three moments about 16 are 0.35, 2.9 and 1.93.

Step 1: (i) Bowley's coefficient = (28.29 + 13.73 - 2 times 26.01) divided by (28.29 - 13.73) = -10/14.56 = -0.687, so negatively skewed.

Step 2: (ii) Mean 140 is less than mode 148.7, so negatively skewed.

Step 3: (iii) Third central moment = 1.93 - 3 times 0.35 times 2.9 + 2 times (0.35)^3 = 1.93 - 3.045 + 0.086 = -1.029, which is negative, so negatively skewed. (The printed key gives positively skewed here, but the third central moment computed from the given moments is negative.)

Answer: (i) Negatively skewed, (ii) negatively skewed, (iii) negatively skewed.

4.35 (i) The second moments about the mean of two distributions are 13.76 and 63.0, and their fourth moments are 528.06 and 9500. Which distribution is (a) leptokurtic, (b) mesokurtic, (c) platykurtic? (ii) The fourth central moment of a symmetrical distribution is 243. What standard deviation makes it mesokurtic?

Step 1: (i) First distribution: b2 = 528.06 divided by (13.76)^2 = 2.79, less than 3. Second distribution: b2 = 9500 divided by 63^2 = 2.39, less than 3. So both distributions are platykurtic; neither is leptokurtic or mesokurtic.

Step 2: (ii) For a mesokurtic distribution b2 = 3, so 243 = 3 times (standard deviation)^4, giving (standard deviation)^4 = 81 and standard deviation = 3.

Answer: (i) Both platykurtic; (ii) S.D. = 3.

4.36 The second moment about the mean is 25. What would the fourth moment be if the distribution is (i) leptokurtic, (ii) mesokurtic, (iii) platykurtic?

Step 1: For a mesokurtic distribution b2 = 3, so the fourth moment = 3 times 25^2 = 1875.

Step 2: For a leptokurtic distribution b2 is greater than 3, so the fourth moment is greater than 1875.

Step 3: For a platykurtic distribution b2 is less than 3, so the fourth moment is less than 1875.

Answer: (i) More than 1875, (ii) 1875, (iii) less than 1875. (The printed key gives only the single value 1875, which is the mesokurtic value.)

4.37 Which of the following is correct for a negatively skewed distribution? (i) A.M. greater than mode, (ii) A.M. less than mode, (iii) A.M. greater than median.

For a negatively skewed distribution the mean is less than the median, which is less than the mode. So the correct option is (ii) A.M. less than mode. (The printed key classifies the options instead: (i) A.M. greater than mode is positively skewed, (ii) A.M. less than mode is negatively skewed, (iii) A.M. greater than median is positively skewed.)

4.38 State the shape or name of the distribution if: (i) Mean = Median = Mode, (ii) Mean > Median > Mode, (iii) Mean < Median < Mode, (iv) b1 = 0 and b2 = 3, (v) b1 = 0 and b2 = 5.

(i) Mean = Median = Mode: the distribution is symmetric. (ii) Mean > Median > Mode: the distribution is positively skewed. (iii) Mean < Median < Mode: the distribution is negatively skewed. (iv) b1 = 0 and b2 = 3: the distribution is symmetric and mesokurtic, like the normal distribution. (v) b1 = 0 and b2 = 5: the distribution is symmetric (b1 = 0) but leptokurtic, since b2 = 5 is greater than 3.

Exercise 4 - 4.39 True or False

4.39 (i) The sum of squares of deviations from the median is minimum.

False

4.39 (ii) The sum of absolute deviations from the mean is minimum.

False

4.39 (iii) If each observation is multiplied by a constant, the variance increases.

True

4.39 (iv) If a constant is added to each observation, the variance increases.

False

4.39 (v) The standard deviation is the positive square root of the variance.

True

4.39 (vi) Range is a measure of absolute dispersion.

True

4.39 (vii) A relative measure is independent of the unit of measurement.

True

4.39 (viii) Mean deviation is not based on all observations.

False

4.39 (ix) Semi-interquartile range is also called interquartile range.

False

4.39 (x) The standard deviation is dependent upon the origin.

False

4.39 (xi) The coefficient of variation is a relative measure of dispersion.

True

4.39 (xii) The first moment about the mean is one.

False

4.39 (xiii) If b1 = 3, the distribution will be symmetrical.

False

4.39 (xiv) Mean deviation is always less than the standard deviation.

True

4.39 (xv) The normal distribution is also known as a mesokurtic distribution.

True

Exercise 4 - 4.40 Fill in the Blanks

4.40 (i) A measure of dispersion is called scatter .

4.40 (ii) A measure of dispersion expressed as a coefficient is called relative measures.

4.40 (iii) The sum of absolute deviations is minimum if computed from the median .

4.40 (iv) The standard deviation does not change if a constant is added to or subtracted from all observations.

4.40 (v) The coefficient of variation is always _____ from the unit of _____.

The coefficient of variation is always free from the unit of measurement.

4.40 (vi) Data having the least coefficient of variation is considered more consistent .

4.40 (vii) Lack of symmetry is called skewness .

4.40 (viii) In a symmetrical distribution, the quartile deviation is equidistant from the median on both sides.

4.40 (ix) Shepherd correction is applicable when the frequency distribution tends to zero in both directions.

4.40 (x) A relative measure is the ratio between the absolute dispersion and the average.

Chapter 5: Index Numbers

Exercise 5 - Questions 5.1 to 5.28

5.1 Define an index number and state its uses.

An index number is a statistical device that measures the relative change in a variable, or in a group of related variables, over time or between places. It is expressed as a percentage of a base-period value.

Its uses include: measuring changes in the general price level; measuring changes in the cost of living; comparing production, trade and employment over time; adjusting wages, salaries and contracts for inflation; and forecasting business conditions.

5.2 Define a price index number and state the main points to be considered in its construction.

A price index number measures the average change in the prices of a commodity or a group of commodities between a base period and a current period.

The main points in constructing price index numbers are: the purpose of the index; the selection of a suitable base period; the selection of representative commodities; the collection of price quotations; the choice of a suitable average; and the system of weights, if any.

5.3 Define an index number and state its different types.

Index numbers are of several types. By the variable measured: price index numbers, quantity (or volume) index numbers and value index numbers. By construction: simple index numbers (one commodity) and composite index numbers (a group of commodities). By base: fixed-base index numbers and chain-base index numbers. By scope: wholesale price index numbers, retail price index numbers and consumer price index numbers.

5.4 What are the important problems in the construction of price index numbers?

The important problems in constructing price index numbers are: choosing a normal base period; selecting representative commodities and keeping their quality constant; obtaining reliable and comparable price quotations; choosing a suitable average (arithmetic mean, geometric mean or median); deciding the weights to be

given to different commodities; and choosing a suitable formula (Laspeyres, Paasche or Fisher).

5.5 Compare: (i) simple and composite index numbers, (ii) fixed-base and chain-base methods.

(i) A simple index number relates to a single commodity, while a composite index number relates to a group of commodities combined into one number.

(ii) In the fixed-base method the same base year is used for all years, while in the chain-base method each year is compared with the immediately preceding year and the resulting link relatives are chained together.

5.6 What is a weighted index number?

A weighted index number is an index number in which each commodity is given a weight according to its relative importance, usually its quantity consumed or its value, so that more important commodities influence the index more than less important ones.

5.7 Calculate the index number of price, taking the average price of all years as base. Years 1970-1977, Prices 15, 19, 21, 30, 37, 38, 40, 48.

Step 1: Average price = (15 + 19 + 21 + 30 + 37 + 38 + 40 + 48) divided by 8 = 248/8 = 31.

Step 2: Index for each year = (price divided by 31) times 100. This gives: 1970: 48.39; 1971: 61.29; 1972: 67.74; 1973: 96.77; 1974: 119.35; 1975: 122.58; 1976: 129.03; 1977: 154.84.

Answer: 48.39, 61.29, 67.74, 96.77, 119.35, 122.58, 129.03, 154.84.

5.8 The prices per maund of a commodity for the years 1945 to 1959 are: 1945: 12.7, 1946: 18.97, 1947: 19.70, 1948: 13.50, 1949: 15.65, 1950: 21.85, 1951:

20.90, 1952: 19.80, 1953: 23.65, 1954: 24.55, 1955: 15.65, 1956: 16.15, 1957: 20.20, 1958: 25.25, 1959: 32.40. Construct index numbers to two decimal places: (i) taking 1945 as base, (ii) taking the average price of all years as base.

Step 1: Note: the question says '1945 to 1960', but the table ends at 1959 and contains 15 values; the solution uses the 15 printed years. The 1950 price is 21.85: the printed key's 1950 index on the 1945 base is 172.05, and 21.85/12.7 times 100 = 172.05, which fixes the price as 21.85.

Step 2: (i) With 1945 (12.7) as base, index = (price/12.7) times 100: 100, 149.37, 155.12, 106.30, 123.23, 172.05, 164.57, 155.91, 186.22, 193.31, 123.23, 127.17, 159.06, 198.82, 255.12. The printed key shows 164.057 for 1951, a misprint for 164.57.

Step 3: (ii) Average price = 300.92/15 = 20.0613. Index = (price/20.0613) times 100: 63.31, 94.56, 98.20, 67.29, 78.01, 108.92, 104.18, 98.70, 117.89, 122.37, 78.01, 80.50, 100.69, 125.86, 161.50. The key differs only in the last decimal places (94.57, 67.30, 78.08, 104.19, 122.38, 78.02, 80.51, 100.70, 125.87, 161.52).

Answer: (i) 100, 149.37, 155.12, 106.30, 123.23, 172.05, 164.57, 155.91, 186.22, 193.31, 123.23, 127.17, 159.06, 198.82, 255.12. (ii) 63.31, 94.56, 98.20, 67.29, 78.01, 108.92, 104.18, 98.70, 117.89, 122.37, 78.01, 80.50, 100.69, 125.86, 161.50.

5.9 Construct index numbers: (i) taking 1977 as base, (ii) taking the average price as base. Prices: 1977: 22.5, 1978: 25, 1979: 27.5, 1980: 30, 1981: 35, 1982: 32.5, 1983: 37.5, 1984: 47.5, 1985: 45.

Step 1: (i) With 1977 (22.5) as base: 100, 111.11, 122.22, 133.33, 155.56, 144.44,

166.67, 211.11, 200.

Step 2: (ii) Average price = 302.5/9 = 33.611. Index = (price/33.611) times 100: 66.94, 74.38, 81.82, 89.26, 104.13, 96.69, 111.57, 141.32, 133.88. The key gives 104.14, 96.70, 141.33 and 133.89 for the fifth, sixth, eighth and ninth values, last-digit rounding differences.

Answer: (i) 100, 111.11, 122.22, 133.33, 155.56, 144.44, 166.67, 211.11, 200. (ii) 66.94, 74.38, 81.82, 89.26, 104.13, 96.69, 111.57, 141.32, 133.88.

5.10 Construct index numbers: (i) taking 1930 as base, (ii) taking the average of the first three years as base, (iii) taking 1935 as base. Prices: 1930: 4, 1931: 5, 1932: 6, 1933: 7, 1934: 8, 1935: 10, 1936: 9, 1937: 10, 1938: 11.

Step 1: (i) With 1930 (4) as base: 100, 125, 150, 175, 200, 250, 225, 250, 275.

Step 2: (ii) Average of the first three years = (4 + 5 + 6)/3 = 5. Index = (price/5) times 100: 80, 100, 120, 140, 160, 200, 180, 200, 220.

Step 3: (iii) With 1935 (10) as base: 40, 50, 60, 70, 80, 100, 90, 100, 110.

Answer: (i) 100, 125, 150, 175, 200, 250, 225, 250, 275. (ii) 80, 100, 120, 140, 160, 200, 180, 200, 220. (iii) 40, 50, 60, 70, 80, 100, 90, 100, 110.

5.11 The wholesale prices of refined petroleum per gallon in the U.K. are, with 1923 as base 100: 1923: 13, 1924: 13 and a half, 1925: 13, 1926: 11 and a half, 1927: 12 and a quarter, 1928: 11 and a quarter, 1929: 10 and a half, 1930: 12 and a half, 1931: 10 and a quarter, 1932: 10 and a half. Construct the series of price relatives.

Step 1: Note on the 1927 price: the fraction in the printed table is unclear in the scan (it reads like 3/7 or 7/8), but the printed key's index for 1927 is 94.23, and 12.25/13 times 100 = 94.23 exactly, which fixes the intended price as 12 and a quarter. The solution uses 12.25.

Step 2: Price relative = (price/13) times 100: 1923: 100; 1924: 103.85; 1925: 100; 1926: 88.46; 1927: 94.23; 1928: 86.54; 1929: 80.77; 1930: 96.15; 1931: 78.85; 1932: 80.77. The key gives 78.84 for 1931, a last-digit slip for 78.85.

Answer: 100, 103.85, 100, 88.46, 94.23, 86.54, 80.77, 96.15, 78.85, 80.77.

5.12 Construct index numbers of prices for the following data taking 1960 as base: 1960: 50, 1961: 52, 1962: 55, 1963: 57, 1964: 62, 1965: 72, 1966: 73, 1967: 75, 1968: 71, 1969: 70.

Step 1: Index = (price/50) times 100: 100, 104, 110, 114, 124, 144, 146, 150, 142, 140.

Answer: 100, 104, 110, 114, 124, 144, 146, 150, 142, 140.

5.13 The prices in Rs. per maund of coal sold during the years 1953-58 are: 1953: 14.95, 1954: 14.95, 1955: 15.10, 1956: 15.65, 1957: 16.28, 1958: 16.28. Compute the index number of price for the year 1953 as base.

Step 1: Index = (price/14.95) times 100: 1953: 100; 1954: 100; 1955: 101.00; 1956: 104.68; 1957: 108.90; 1958: 108.90.

Step 2: Note: the printed table gives the 1958 price as 16.28, the same as 1957, so the 1958 index is 108.90. The printed key gives 100.58 for 1958, which would need

a 1958 price of about 15.03; that value is not in the table, so the key's last entry is treated as a misprint. The key also gives 101.07 for 1955 against the computed 101.00.

Answer: 100, 100, 101.00, 104.68, 108.90, 108.90.

5.14 From the data given below, compute the index numbers of prices, taking 1962 as base. 1962: Firewood 3.25, Soft cake 2.50, Kerosene oil 0.20, Matches 0.06. 1963: 3.44, 2.80, 0.22, 0.06. 1964: 3.50, 2.00, 0.25, 0.06. 1965: 3.75, 2.50, 0.25, 0.06.

Step 1: Method used: simple average of price relatives, stated here because the question does not name a method.

Step 2: Price relatives (price divided by the 1962 price, times 100). 1963: Firewood 105.85, Soft cake 112, Kerosene oil 110, Matches 100; average = 106.96. 1964: 107.69, 80, 125, 100; average = 103.17. 1965: 115.38, 100, 125, 100; average = 110.10.

Answer: 1962: 100; 1963: 106.96; 1964: 103.17; 1965: 110.10 (the key gives 103.175 and 110.095, the same values to more decimal places).

5.15 Compute index numbers of prices from the following data taking 1981 as base and using median as average. 1981: A 18, B 85, C 52. 1982: 22, 76, 60. 1983: 28, 80, 66. 1984: 31, 95, 80.

Step 1: Price relatives (price divided by the 1981 price, times 100). 1982: A 122.22, B 89.41, C 115.38. 1983: A 155.56, B 94.12, C 126.92. 1984: A 172.22, B 111.76, C 153.85.

Step 2: The arithmetic means of the relatives are: 1982: (122.22 + 89.41 + 115.38)/3 = 109.01; 1983: (155.56 + 94.12 + 126.92)/3 = 125.53; 1984: (172.22 + 111.76 + 153.85)/3 = 145.94. These are exactly the printed key's values.

Step 3: Note: the medians of the relatives would be 115.38, 126.92 and 153.85, which do not match the key. The question's word 'median' is therefore treated as a misprint for 'mean', and the solution uses the arithmetic mean, following the key.

Answer: 1981: 100; 1982: 109.01; 1983: 125.53; 1984: 145.94.

5.16 Find chain index numbers (using geometric mean to average the relatives) for the following data of prices, taking 1970 as the base year. A: 1970: 40, 1971: 43, 1972: 45, 1973: 42, 1974: 50. B: 160, 162, 165, 161, 168. C: 20, 29, 52, 23, 27. D: 240, 245, 247, 250, 255.

Step 1: Link relatives (each year's price divided by the preceding year's price, times 100). 1971: A 107.50, B 101.25, C 145.00, D 102.08; geometric mean = 112.66. Chain index for 1971 = 112.66.

Step 2: 1972 link relatives: A 104.65, B 101.85, C 179.31, D 100.82; geometric mean = 117.82. Chain index for 1972 = 112.66 times 117.82/100 = 132.74.

Step 3: 1973 link relatives: A 93.33, B 97.58, C 44.23, D 101.21; geometric mean = 79.91. Chain index for 1973 = 132.74 times 79.91/100 = 106.07. The fall is caused by commodity C, whose price drops from 52 to 23.

Step 4: 1974 link relatives: A 119.05, B 104.35, C 117.39, D 102.00; geometric mean = 110.44. Chain index for 1974 = 106.07 times 110.44/100 = 117.14.

Step 5: Note: the printed key gives 100, 112.66, 132.49, 136.20, 146.74. It agrees at 1971, differs by 0.25 at 1972, and its 1973-74 values do not follow from the printed table: they correspond to C prices of 63 and 67 for 1973 and 1974, while the table prints 23 and 27. The chain above follows the printed table.

Answer: 1970: 100; 1971: 112.66; 1972: 132.74; 1973: 106.07; 1974: 117.14.

5.17 The following table gives the average wholesale prices in rupees per unit of gold, wheat and cotton during the years 1912-1917. Construct index numbers with 1912 as base using (i) A.M. (ii) G.M. Gold: 1912: 25.3, 1913: 30.8, 1914: 33.4, 1915: 35.5, 1916: 35.4, 1917: 36.0. Wheat: 17.3, 14.5, 4.9, 5.7, 17.7, 11.6. Cotton: 7.8, 5.4, 6.7, 5.6, 7.2, 10.2.

Step 1: Price relatives (price divided by the 1912 price, times 100). Gold: 100, 121.74, 132.02, 140.32, 139.92, 142.29. Wheat: 100, 83.82, 28.32, 32.95, 102.31, 67.05. Cotton: 100, 69.23, 85.90, 71.79, 92.31, 130.77.

Step 2: (i) Arithmetic means of the relatives: 1912: 100; 1913: 91.59; 1914: 82.08; 1915: 81.69; 1916: 111.51; 1917: 113.37.

Step 3: (ii) Geometric means of the relatives: 1912: 100; 1913: 89.06; 1914: 68.48; 1915: 69.24; 1916: 109.74; 1917: 107.65.

Step 4: Note: the printed key agrees with the computed series except for 1916, where it gives A.M. 115.57 and G.M. 114.32 against the computed 111.51 and 109.74. The 1916 prices are printed clearly (gold 35.4, wheat 17.7, cotton 7.2), and no single price change reproduces the key's 1916 values exactly, so the key's 1916 entries are treated as misprints. The key also differs slightly at 1915 (A.M. 81.55,

G.M. 69.17).

Answer: (i) A.M.: 100, 91.59, 82.08, 81.69, 111.51, 113.37. (ii) G.M.: 100, 89.06, 68.48, 69.24, 109.74, 107.65.

5.18 Construct chain indices for the following years, taking 1940 as base. Wheat: 1940: 2.80, 1941: 3.40, 1942: 3.60, 1943: 4.00, 1944: 4.20. Rice: 2.95, 3.60, 2.90, 2.75, 2.75. Maize: 3.10, 3.50, 3.40, 4.50, 3.70.

Step 1: Method used: arithmetic mean of link relatives, stated here because the question does not name an average.

Step 2: Link relatives (each year's price divided by the preceding year's price, times 100). 1941: Wheat 121.43, Rice 122.03, Maize 112.90; average = 118.79. Chain index for 1941 = 118.8.

Step 3: 1942 link relatives: Wheat 105.88, Rice 80.56, Maize 97.14; average = 94.53. Chain index for 1942 = 118.8 times 94.53/100 = 112.3.

Step 4: 1943 link relatives: Wheat 111.11, Rice 94.83, Maize 132.35; average = 112.76. Chain index for 1943 = 112.3 times 112.76/100 = 126.6.

Step 5: 1944 link relatives: Wheat 105.00, Rice 100.00, Maize 82.22; average = 95.74. Chain index for 1944 = 126.6 times 95.74/100 = 121.2.

Step 6: Note: the printed key gives 100, 114.3, 119.4, 128.7, 134.9, which follows from the printed table under neither the arithmetic-mean chain (100, 118.8, 112.3, 126.6, 121.2) nor the geometric-mean chain (100, 118.7, 111.5, 124.6, 118.6). The

key's series is therefore treated as inconsistent with the printed data, and the arithmetic-mean chain above is given.

Answer: 1940: 100; 1941: 118.8; 1942: 112.3; 1943: 126.6; 1944: 121.2.

5.19 Construct index numbers for 1963 assuming 1953 as base period by (i) Laspeyre's formula (ii) Paasche's formula. A: 1953 Price 2 Quantity 50, 1963 Price 10 Quantity 40. B: 3 and 10, 8 and 5. C: 4 and 5, 4 and 5.

Step 1: (i) Laspeyres = (sum of current price times base quantity) divided by (sum of base price times base quantity), times 100 = (10 times 50 + 8 times 10 + 4 times 5) divided by (2 times 50 + 3 times 10 + 4 times 5), times 100 = 600/150 times 100 = 400.

Step 2: (ii) Paasche = (sum of current price times current quantity) divided by (sum of base price times current quantity), times 100 = (10 times 40 + 8 times 5 + 4 times 5) divided by (2 times 40 + 3 times 5 + 4 times 5), times 100 = 460/115 times 100 = 400.

Answer: (i) Laspeyres 400, (ii) Paasche 400.

5.20 Compute the weighted index numbers for 1964 from the following data with 1960 as base. Milk: Price 1960: 3.95, 1964: 4.25, Quantity 1960: 97.75, 1964: 104.36. Cheese: 34.80, 38.90, 78, 83. Butter: 61.56, 59.70, 118, 116.

Step 1: Method used: Laspeyres (base-year weights), stated here because the question does not name a formula. It is the method that gives the printed key's value 101.2508.

Step 2: Sum of current price times base quantity = 4.25 times 97.75 + 38.90 times 78 + 59.70 times 118 = 415.44 + 3034.20 + 7044.60 = 10494.24.

Step 3: Sum of base price times base quantity = 3.95 times 97.75 + 34.80 times 78 + 61.56 times 118 = 386.11 + 2714.40 + 7264.08 = 10364.59.

Step 4: Laspeyres index = 10494.24/10364.59 times 100 = 101.2508. For reference, Paasche = 101.4926 and Fisher = 101.3716.

Answer: 101.2508 (Laspeyres).

5.21 Compute Fisher's index number for the following data. Base Year: A Price 7 Quantity 70, B 5 and 27, C 10 and 35, D 9 and 50, E 3 and 16. Current Year: A 5 and 49, B 7 and 28, C 9 and 29, D 4 and 42, E 10 and 25.

Step 1: Sum of base price times base quantity = 7 times 70 + 5 times 27 + 10 times 35 + 9 times 50 + 3 times 16 = 490 + 135 + 350 + 450 + 48 = 1473.

Step 2: Sum of current price times base quantity = 5 times 70 + 7 times 27 + 9 times 35 + 4 times 50 + 10 times 16 = 350 + 189 + 315 + 200 + 160 = 1214. Laspeyres = 1214/1473 times 100 = 82.42.

Step 3: Sum of current price times current quantity = 5 times 49 + 7 times 28 + 9 times 29 + 4 times 42 + 10 times 25 = 245 + 196 + 261 + 168 + 250 = 1120. Sum of base price times current quantity = 7 times 49 + 5 times 28 + 10 times 29 + 9 times 42 + 3 times 25 = 343 + 140 + 290 + 378 + 75 = 1226. Paasche = 1120/1226 times 100 = 91.35.

Step 4: Fisher = square root of (Laspeyres times Paasche) = square root of (82.42 times 91.35) = 86.77.

Answer: 86.77.

5.22 Construct Fisher's ideal index taking 1970 as base. Total Production (tons) and Harvest Price (Rs): Rice 1970: 71 tons, Rs. 3.80; 1971: 26 tons, Rs. 3.50. Barley 107, 2.90; 83, 1.90. Maize 72, 2.90; 48, 1.80.

Step 1: Sum of base price times base quantity = 3.80 times 71 + 2.90 times 107 + 2.90 times 72 = 269.80 + 310.30 + 208.80 = 788.90.

Step 2: Sum of current price times base quantity = 3.50 times 71 + 1.90 times 107 + 1.80 times 72 = 248.50 + 203.30 + 129.60 = 581.40. Laspeyres = 581.40/788.90 times 100 = 73.70.

Step 3: Sum of current price times current quantity = 3.50 times 26 + 1.90 times 83 + 1.80 times 48 = 91.00 + 157.70 + 86.40 = 335.10. Sum of base price times current quantity = 3.80 times 26 + 2.90 times 83 + 2.90 times 48 = 98.80 + 240.70 + 139.20 = 478.70. Paasche = 335.10/478.70 times 100 = 70.00.

Step 4: Fisher = square root of (73.70 times 70.00) = 71.83.

Answer: 71.83.

5.23 Calculate Fisher's Ideal index from the following data with 1965 as base year. A: 1965 Price 4.6 Quantity 102, 1970 Price 9.50 Quantity 96. B: 3.7 and 15, 7.36 and 28. C: 10.2 and 17, 8.42 and 21. D: 8.9 and 19, 9.87 and 13.

Step 1: Sum of base price times base quantity = 4.6 times 102 + 3.7 times 15 + 10.2 times 17 + 8.9 times 19 = 469.20 + 55.50 + 173.40 + 169.10 = 867.20.

Step 2: Sum of current price times base quantity = 9.50 times 102 + 7.36 times 15 + 8.42 times 17 + 9.87 times 19 = 969.00 + 110.40 + 143.14 + 187.53 = 1410.07. Laspeyres = 1410.07/867.20 times 100 = 162.60.

Step 3: Sum of current price times current quantity = 9.50 times 96 + 7.36 times 28 + 8.42 times 21 + 9.87 times 13 = 912.00 + 206.08 + 176.82 + 128.31 = 1423.21. Sum of base price times current quantity = 4.6 times 96 + 3.7 times 28 + 10.2 times 21 + 8.9 times 13 = 441.60 + 103.60 + 214.20 + 115.70 = 875.10. Paasche = 1423.21/875.10 times 100 = 162.63.

Step 4: Fisher = square root of (162.60 times 162.63) = 162.62.

Answer: 162.62.

5.24 Define weighted and unweighted index numbers and explain why weighted index numbers are preferred over unweighted index numbers.

An unweighted index number gives equal importance to every commodity, while a weighted index number gives each commodity a weight according to its relative importance.

Weighted index numbers are preferred because commodities are not equally important in practice: a large change in the price of a necessity should affect the index more than the same change in the price of a minor item. An unweighted index overstates unimportant items and understates important ones, and can therefore mislead.

5.25 Calculate Laspeyre's, Paasche's and Fisher's ideal index for the following data with 1992 as base. Wheat flour: Price 1992: 4.38, 1993: 4.57, Quantity 20 Kg, 16 Kg. Rice: 14.15, 15.58, 10 Kg, 12 Kg. Moong pulse: 18.67, 17.28, 1 Kg, 1 Kg. Gram pulse: 10.41, 16.36, 1 Kg, 1 Kg.

Step 1: Sum of base price times base quantity = 4.38 times 20 + 14.15 times 10 + 18.67 times 1 + 10.41 times 1 = 87.60 + 141.50 + 18.67 + 10.41 = 258.18.

Step 2: Sum of current price times base quantity = 4.57 times 20 + 15.58 times 10 + 17.28 times 1 + 16.36 times 1 = 91.40 + 155.80 + 17.28 + 16.36 = 280.84. Laspeyres = 280.84/258.18 times 100 = 108.78.

Step 3: Sum of current price times current quantity = 4.57 times 16 + 15.58 times 12 + 17.28 times 1 + 16.36 times 1 = 73.12 + 186.96 + 17.28 + 16.36 = 293.72. Sum of base price times current quantity = 4.38 times 16 + 14.15 times 12 + 18.67 times 1 + 10.41 times 1 = 70.08 + 169.80 + 18.67 + 10.41 = 268.96. Paasche = 293.72/268.96 times 100 = 109.21.

Step 4: Fisher = square root of (108.78 times 109.21) = 108.99.

Answer: Laspeyres 108.78, Paasche 109.21, Fisher 108.99.

5.26 The following figures give the average annual prices in U.K. for beef and mutton. 1935: Beef 54, Mutton 75. 1936: 54, 73. 1937: 61, 73. 1938: 62, 62. 1939: 61, 68. 1940: 72, 85. 1941: 72, 85. 1942: 76, 90. 1943: 79, 90. Construct the index number of meat prices giving weights 2 and 1 for beef and mutton respectively. Take the year 1935 as the base year.

Step 1: Index for each year = (2 times beef relative + 1 times mutton relative)

divided by 3, where each relative is the year's price divided by the 1935 price, times 100.

Step 2: 1935: 100. 1936: (2 times 100 + 97.33)/3 = 99.11. 1937: (2 times 112.96 + 97.33)/3 = 107.75. 1938: (2 times 114.81 + 82.67)/3 = 104.10. 1939: (2 times 112.96 + 90.67)/3 = 105.53.

Step 3: 1940: (2 times 133.33 + 113.33)/3 = 126.67. 1941: the same prices as 1940, so 126.67. 1942: (2 times 140.74 + 120)/3 = 133.83. 1943: (2 times 146.30 + 120)/3 = 137.53.

Step 4: Note: the printed key gives 100, 99.1, 109.97, 104.1, 105.53, 126.66, 128.93, 122.67, 140.2. It agrees for 1935, 1936, 1938, 1939 and 1940, but its values for 1937, 1941, 1942 and 1943 do not follow from the printed table, so those four key entries are treated as misprints.

Answer: 100, 99.11, 107.75, 104.10, 105.53, 126.67, 126.67, 133.83, 137.53.

5.27 The following table shows the average price in rupees for wheat, rice and barley. 1980: Wheat 175.5, Rice 480.4, Barley 82.4. 1981: 180.3, 509.7, 90.6. Taking 1980 as base year, construct the price index number by the weighted average of relatives method for the year 1981 using the weight 20 for wheat, 12 for rice and 4 for barley.

Step 1: Price relatives for 1981: Wheat = 180.3/175.5 times 100 = 102.74; Rice = 509.7/480.4 times 100 = 106.10; Barley = 90.6/82.4 times 100 = 109.95.

Step 2: Weighted average = (20 times 102.74 + 12 times 106.10 + 4 times 109.95)

divided by (20 + 12 + 4) = (2054.70 + 1273.19 + 439.81) divided by 36 = 3767.70/36 = 104.66.

Step 3: Note: the printed key gives 73.002, which cannot be obtained from these prices and weights by any standard averaging (every relative exceeds 100, so any weighted average must exceed 100). The key's value is treated as a misprint.

Answer: 104.66.

5.28 An inquiry into the budgets of middle class families in a city of England gave the following information. What change in cost of living do the figures of 1929 show as compared with 1928? Expenses shares: Food 35 percent, Rent 15 percent, Clothing 20 percent, Fuel 10 percent, Misc. 20 percent. Price (1928): Food 150, Rent 30, Clothing 75, Fuel 25, Misc. 40. Price (1929): 145, 30, 65, 23, 45.

Step 1: Price relatives for 1929 (1928 = 100): Food = 145/150 times 100 = 96.67; Rent = 30/30 times 100 = 100; Clothing = 65/75 times 100 = 86.67; Fuel = 23/25 times 100 = 92; Misc. = 45/40 times 100 = 112.50.

Step 2: Weighted average of relatives = (35 times 96.67 + 15 times 100 + 20 times 86.67 + 10 times 92 + 20 times 112.50) divided by 100 = (3383.33 + 1500 + 1733.33 + 920 + 2250) divided by 100 = 9786.67/100 = 97.87.

Step 3: The cost of living in 1929 is therefore 97.87 percent of 1928, a fall of about 2.13 percent. Note: the printed key gives 98.15; no single clean price misread in the table reproduces 98.15, so the key's value is treated as a misprint and the computed 97.87 is given.

Answer: 97.87, a fall of about 2.13 percent from 1928.

Exercise 5 - 5.29 Fill in the Blanks

5.29 (i) Changes in wholesale and retail prices are studied in price-numbers .

5.29 (ii) Volume or quantity changes are compared by volume index numbers .

5.29 (iii) In aggregative index numbers both quantities and prices are used.

5.29 (iv) Index numbers are used for _____ business activity and discovering _____ fluctuation and business _____.

Index numbers are used for forecasting business activity and discovering seasonal fluctuation and business cycles.

5.29 (v) The purpose of an index number may be _____ or _____.

The purpose of an index number may be general or special.

5.29 (vi) The two methods of selection of base periods are the _____ and the _____.

The two methods of selection of base periods are the fixed base method and the chain base method.

5.29 (vii) The base period in the fixed base method should be a normal year .

5.29 (viii) The _____ process is a must in the _____ method for comparison purposes.

The chaining process is a must in the chain base method for comparison purposes.

5.29 (ix) The geometric mean is a suitable average in the chain base method.

5.29 (x) Unweighted indices are classified into simple _____ indices and simple _____.

Unweighted indices are classified into simple aggregative indices and simple average of relatives indices.

Exercise 5 - 5.30 True or False

5.30 (i) Six steps are involved in the construction of price index numbers.

True

5.30 (ii) In a price relative, the given year price is divided by the base year price.

True

5.30 (iii) Laspeyres index is also named the current-year weighted index.

False

5.30 (iv) Fisher's index is the geometric mean of Laspeyres and Paasche indices.

True

5.30 (v) The aggregate expenditure method and the family budget method are types of consumer price index.

True

5.30 (vi) Index numbers are calculated in percentages.

True

5.30 (vii) Index numbers are statistical barometers.

False

5.30 (viii) In the chain base method, the base year is fixed.

False

5.30 (ix) The most suitable average for index numbers is the harmonic mean.

False

5.30 (x) The smaller the sample size, the greater the accuracy.

False

Exercise 5 - Note on the Answers key

Note on 5.30 (vii): the printed key (p. 255) marks this statement False.

The statement in 5.30 (vii) says that index numbers are 'statistical barometers'. The standard phrase is that index numbers are 'economic barometers', because they measure economic conditions. The answer given here follows the printed key, which

marks (vii) False.

Chapter 6: Probability

Exercise 6 - Questions 6.1 to 6.30

6.1 (i) Distinguish between permutation and combination. (ii) If A = {1, 3, 5, 7}, B = {2, 4, 6, 8}, C = {1, 2, 3, 4, 5} and S = {1, 2, ..., 8}, find (a) A intersect B, (b) C union A, (c) the complement of (A intersect C), (d) B intersect C.

(i) A permutation is an orderly arrangement of objects, where the order matters; a combination is a selection of objects, where the order does not matter. For example, from the letters A, B and C, the arrangements AB and BA are two different permutations but the same combination.

(ii) A = {1, 3, 5, 7}, B = {2, 4, 6, 8}, C = {1, 2, 3, 4, 5} and S = {1, 2, ..., 8}. (a) A and B have no element in common, so A intersect B = { } (the empty set). (b) C union A = {1, 2, 3, 4, 5, 7}. (c) A intersect C = {1, 3, 5}, and its complement in S is {2, 4, 6, 7, 8}. (d) B intersect C = {2, 4}.

6.2 Evaluate: (i) 7!, (ii) 16! divided by 8!, (iii) 17! divided by (8! times 4!), (iv) 9P6, (v) 15P6, (vi) 5P5, (vii) 5C3, (viii) 52C13, (ix) 11C4.

Step 1: (i) 7! = 7 times 6 times 5 times 4 times 3 times 2 times 1 = 5040.

Step 2: (ii) 16! divided by 8! = 9 times 10 times 11 times 12 times 13 times 14 times 15 times 16 = 518918400.

Step 3: (iii) 17! divided by (8! times 4!) = 367567200.

Step 4: (iv) 9P6 = 9 times 8 times 7 times 6 times 5 times 4 = 60480.

Step 5: (v) 15P6 = 15 times 14 times 13 times 12 times 11 times 10 = 3603600.

Step 6: (vi) 5P5 = 5! = 120.

Step 7: (vii) 5C3 = 10.

Step 8: (viii) 52C13 = 635013559600.

Step 9: (ix) 11C4 = 330.

Answer: (i) 5040, (ii) 518918400, (iii) 367567200, (iv) 60480, (v) 3603600, (vi) 120, (vii) 10, (viii) 635013559600, (ix) 330.

6.3 If A = {1, 4}, B = {2, 3}, C = {3} and S = {1, 2, 3, 4}, find (i) A times B, (ii) B times A, (iii) A times A, (iv) B times B.

A = {1, 4}, B = {2, 3} and S = {1, 2, 3, 4}. (i) A times B = {(1, 2), (1, 3), (4, 2), (4, 3)}. (ii) B times A = {(2, 1), (2, 4), (3, 1), (3, 4)}. (iii) A times A = {(1, 1), (1, 4), (4, 1), (4, 4)}. (iv) B times B = {(2, 2), (2, 3), (3, 2), (3, 3)}. Note: the set C = {3} is given in the question but none of the four parts asks about C.

6.4 Define: sample space, event, sample point, simple event and compound event.

Sample space: the set of all possible outcomes of a random experiment, for example {H, T} for tossing a coin. Event: any subset of the sample space, for example getting a head. Sample point: a single outcome in the sample space, for example H. Simple event: an event containing exactly one sample point, for example getting a 6 on a die. Compound event: an event containing more than one sample point, for example getting an even number on a die.

6.5 Define mutually exclusive, independent and dependent events, with examples.

Mutually exclusive events: two events that have no sample point in common, so they cannot occur together; for example, getting a 2 and getting a 5 on one throw of a die. Independent events: the occurrence of one does not affect the probability of the other; for example, the outcomes of two separate tosses of a coin. Dependent events: the probability of one event is affected by whether the other has occurred; for example, drawing two cards from a pack without replacing the first.

6.6 Define a random experiment, a sample space and an event, with examples.

Random experiment: an experiment whose outcome cannot be predicted with certainty in advance, although the set of all possible outcomes is known. Example: tossing a coin. Sample space: the set of all possible outcomes, for example {H, T} for one coin and {HH, HT, TH, TT} for two coins. Event: a subset of the sample space, for example getting at least one head when two coins are tossed, which is {HT, TH, HH}.

6.7 State and prove the multiplicative law of probability for two events A and B which are not statistically independent.

For two events A and B that are not statistically independent, the multiplicative law states: P(A and B) = P(A) times P(B given A) = P(B) times P(A given B).

Proof: By the definition of conditional probability, P(B given A) = P(A and B) divided by P(A), provided P(A) is not zero. Multiplying by P(A) gives P(A and B) = P(A) times P(B given A). The second form follows the same way, starting from P(A given B).

6.8 Find the probability of (i) getting a head in tossing a fair coin, (ii) getting a 5 in rolling a six-faced cubical die, (iii) getting an even number with a perfect cubical die.

Step 1: (i) The sample space is {H, T} with 2 equally likely outcomes, so P(head) = 1/2.

Step 2: (ii) The die has 6 equally likely faces, so P(5) = 1/6.

Step 3: (iii) The even faces are 2, 4 and 6, that is 3 out of 6 outcomes, so P(even) = 3/6 = 1/2.

Answer: (i) 1/2, (ii) 1/6, (iii) 1/2.

6.9 Find the probability of drawing a diamond from a pack of 52 cards.

There are 13 diamonds in a pack of 52 cards, so the probability of drawing a diamond is 13/52 = 1/4.

6.10 Show that in a single throw with two dice, the chance of getting a sum greater than 7 is equal to the chance of getting a sum less than 7.

With two dice there are 36 equally likely outcomes. Sums less than 7 are 2, 3, 4, 5 and 6, occurring in 1 + 2 + 3 + 4 + 5 = 15 ways. Sums greater than 7 are 8, 9, 10, 11 and 12, occurring in 5 + 4 + 3 + 2 + 1 = 15 ways. Hence P(sum less than 7) = 15/36 and P(sum greater than 7) = 15/36, so the two chances are equal.

6.11 A bag contains 12 balls, 3 of which are marked. If 5 balls are drawn together, find the probability that all 3 marked balls are among the 5 drawn.

Step 1: Total selections of 5 balls from 12 = C(12, 5) = 792.

Step 2: Favourable selections = choose all 3 marked balls and any 2 of the other 9 balls = C(3, 3) times C(9, 2) = 1 times 36 = 36.

Step 3: Probability = 36/792 = 1/22.

Answer: 1/22.

6.12 Find the probability of throwing either 7 or more than 10 with two dice.

Step 1: P(sum 7) = 6/36, since 7 occurs in 6 ways.

Step 2: P(sum more than 10) = P(11) + P(12) = 2/36 + 1/36 = 3/36.

Step 3: The two events are mutually exclusive, so P(7 or more than 10) = 6/36 + 3/36 = 9/36 = 1/4.

Answer: 1/4.

6.13 A bag contains 2 red, 3 green, 5 blue and 2 yellow balls. Find the probability that in a sample of 4 balls, balls of all colours are represented.

Step 1: Total selections of 4 balls from 12 = C(12, 4) = 495.

Step 2: For all four colours to appear in 4 balls, exactly one ball of each colour must be drawn: 2 times 3 times 5 times 2 = 60 favourable selections.

Step 3: Probability = 60/495 = 4/33.

Answer: 4/33.

6.14 A bag contains 5 white and 7 black balls. Three balls are drawn. Find the probability of (i) all white, (ii) two white and one black, (iii) all of the same colour.

Step 1: Total selections of 3 balls from 12 = C(12, 3) = 220.

Step 2: (i) P(all white) = C(5, 3) divided by 220 = 10/220 = 1/22.

Step 3: (ii) P(two white, one black) = C(5, 2) times C(7, 1) divided by 220 = 70/220 = 7/22.

Step 4: (iii) P(all same colour) = (C(5, 3) + C(7, 3)) divided by 220 = (10 + 35)/220 = 45/220 = 9/44.

Answer: (i) 1/22, (ii) 7/22, (iii) 9/44.

6.15 Find (i) the probability of getting a sum of 8 in a single toss of a pair of fair dice, (ii) the probability of getting a sum of 7 or 11, (iii) the probability that a ball drawn from a bag of 5 red, 6 white, 4 blue and 3 orange balls is red or blue.

Step 1: (i) A sum of 8 occurs in 5 ways out of 36, so P = 5/36.

Step 2: (ii) A sum of 7 occurs in 6 ways and a sum of 11 in 2 ways, so P = (6 + 2)/36 = 8/36.

Step 3: (iii) The bag has 18 balls, of which 5 + 4 = 9 are red or blue, so P = 9/18.

Answer: (i) 5/36, (ii) 8/36, (iii) 9/18.

6.16 Two cards are drawn from a pack of 52. Find the probability that they are (i) one king and the other a queen, (ii) both of the same colour, (iii) both of different colours.

Step 1: Total selections of 2 cards from 52 = C(52, 2) = 1326.

Step 2: (i) One king and one queen: C(4, 1) times C(4, 1) = 16, so P = 16/1326.

Step 3: (ii) Both same colour: (C(26, 2) + C(26, 2)) divided by 1326 = 650/1326.

Step 4: (iii) Both different colours: 26 times 26 divided by 1326 = 676/1326.

Answer: (i) 16/1326, (ii) 650/1326, (iii) 676/1326.

6.17 A bag contains 9 white and 12 black balls. Find the probability of drawing 5 black balls from the 21 balls.

Step 1: Total selections of 5 balls from 21 = C(21, 5) = 20349.

Step 2: Favourable selections = C(12, 5) = 792.

Step 3: Probability = 792/20349.

Answer: 792/20349.

6.18 A bag contains 5 white and 3 black balls. Two balls are drawn. Find the probability that both are of the same colour.

Step 1: Total selections of 2 balls from 8 = C(8, 2) = 28.

Step 2: Favourable selections = C(5, 2) + C(3, 2) = 10 + 3 = 13.

Step 3: Probability = 13/28.

Answer: 13/28.

6.19 A set of 8 cards includes one joker. A chooses 5 cards and B takes the remaining 3. Find the probability that A has the joker.

A has 5 cards and B has the remaining 3, so A holds the joker with probability 5/8, since the joker is equally likely to be in any of the 8 positions.

6.20 Two cards are drawn from a pack of 52. Find the probability that one is a king

and the other a queen.

This is the same calculation as Question 6.16 (i): one king and one queen can be chosen in C(4, 1) times C(4, 1) = 16 ways out of C(52, 2) = 1326 selections, so the probability is 16/1326.

6.21 Find the probability that a poker hand of 5 cards contains (i) 2 aces and 2 kings, (ii) 5 spades.

Step 1: Total poker hands = C(52, 5) = 2598960.

Step 2: (i) Choose 2 aces from 4, 2 kings from 4, and 1 card from the remaining 44: 6 times 6 times 44 = 1584. Probability = 1584/2598960 = 33/54145.

Step 3: (ii) Choose 5 spades from 13: C(13, 5) = 1287. Probability = 1287/2598960 = 33/66640.

Answer: (i) 33/54145, (ii) 33/66640. Note: the printed key gives 33/54145 for part (ii) as well, but that is the value of part (i) copied across; the correct value for 5 spades is 33/66640.

6.22 A bag contains 14 identical balls: 4 red, 5 black and 5 white. Three balls are drawn. Find the probability of (i) 3 red balls, (ii) at least two white balls.

Step 1: Total selections of 3 balls from 14 = C(14, 3) = 364.

Step 2: (i) P(3 red) = C(4, 3) divided by 364 = 4/364.

Step 3: (ii) P(at least two white) = (C(5, 2) times C(9, 1) + C(5, 3)) divided by 364 = (90 + 10)/364 = 100/364.

Answer: (i) 4/364, (ii) 100/364.

6.23 A marble is drawn from a box containing 10 red, 30 white, 20 blue and 15 orange marbles. Find the probability that it is (i) orange or red, (ii) not red or orange, (iii) not blue, (iv) red, white or blue.

Step 1: The box has 10 + 30 + 20 + 15 = 75 marbles.

Step 2: (i) P(orange or red) = (15 + 10)/75 = 25/75.

Step 3: (ii) P(not red or orange) = (30 + 20)/75 = 50/75.

Step 4: (iii) P(not blue) = (75 - 20)/75 = 55/75.

Step 5: (iv) P(red, white or blue) = (10 + 30 + 20)/75 = 60/75.

Answer: (i) 25/75, (ii) 50/75, (iii) 55/75, (iv) 60/75. Note: the book prints the third part as a second '(ii)'; the parts are answered here in order, without renumbering the question.

6.24 Define conditional probability, with an example.

The conditional probability of an event A, given that an event B has occurred, is P(A given B) = P(A and B) divided by P(B), provided P(B) is not zero. For example, if a card drawn from a pack is known to be a face card, the probability that it is a king is 4/12 = 1/3.

6.25 State and prove the multiplication laws of probability for independent and for dependent events.

For independent events A and B, the multiplication law is P(A and B) = P(A) times

P(B). Proof: independence means P(B given A) = P(B), and the general law P(A and B) = P(A) times P(B given A) then gives P(A and B) = P(A) times P(B).

For dependent events, the multiplication law is P(A and B) = P(A) times P(B given A) = P(B) times P(A given B), which follows directly from the definition of conditional probability.

6.26 If A can solve a problem with probability 60 percent and B with probability 80 percent, find the probability that either A or B solves a problem selected at random.

Step 1: P(A solves) = 0.6 and P(B solves) = 0.8. Assuming A and B work independently, P(both solve) = 0.6 times 0.8 = 0.48.

Step 2: P(either solves) = P(A) + P(B) - P(both) = 0.6 + 0.8 - 0.48 = 0.92.

Answer: 0.92.

6.27 A class has 10 men and 20 women. Half of the men and half of the women have brown eyes. Find the probability that a person selected at random is a man or has brown eyes.

Step 1: Total persons = 30. P(man) = 10/30.

Step 2: Persons with brown eyes = half of 10 men + half of 20 women = 5 + 10 = 15, so P(brown eyes) = 15/30.

Step 3: Men with brown eyes = 5, so P(man and brown eyes) = 5/30.

Step 4: P(man or brown eyes) = 10/30 + 15/30 - 5/30 = 20/30 = 2/3.

Answer: 2/3.

6.28 A box contains 9 tickets numbered 1 to 9. Three tickets are drawn one at a time. Find the probability that they are drawn alternately odd, even, odd or even, odd, even.

Step 1: There are 5 odd and 4 even tickets.

Step 2: P(odd, even, odd) = (5/9) times (4/8) times (4/7) = 80/504.

Step 3: P(even, odd, even) = (4/9) times (5/8) times (3/7) = 60/504.

Step 4: The two patterns are mutually exclusive, so the required probability = (80 + 60)/504 = 140/504 = 5/18.

Answer: 5/18.

6.29 Two drawings of 3 balls each are made from a bag of 5 white and 8 black balls, without replacing the first drawing before the second trial. Find the probability that the first drawing gives 3 white balls and the second gives 3 black balls.

Step 1: First drawing: P(3 white) = C(5, 3) divided by C(13, 3) = 10/286 = 5/143.

Step 2: After the first drawing, the bag has 2 white and 8 black balls left. Second drawing: P(3 black) = C(8, 3) divided by C(10, 3) = 56/120 = 7/15.

Step 3: Required probability = (5/143) times (7/15) = 35/2145 = 7/429.

Answer: 7/429.

6.30 Three balls are drawn successively from a box containing 6 red, 4 white and 5

blue balls, in the order red, white, blue, if each ball is (i) replaced, (ii) not replaced.

Step 1: (i) With replacement, each draw is from 15 balls: P = (6/15) times (4/15) times (5/15) = 120/3375 = 8/225 = 0.0356 (the key rounds this to 0.0355).

Step 2: (ii) Without replacement: P = (6/15) times (4/14) times (5/13) = 120/2730 = 4/91 = 0.0440 (the key rounds this to 0.04395).

Answer: (i) 0.0356, (ii) 0.0440.

Exercise 6 - Questions 6.31 to 6.53

6.31 Bag 1 contains 4 white and 2 black balls; bag 2 contains 3 white and 5 black balls. One ball is drawn from each bag. Find the probability that (i) both are white, (ii) both are black.

Step 1: (i) P(both white) = (4/6) times (3/8) = 12/48 = 0.25.

Step 2: (ii) P(both black) = (2/6) times (5/8) = 10/48 = 0.2083.

Answer: (i) 0.25, (ii) 0.2083.

6.32 Two urns contain respectively 3 white, 7 red and 15 black balls, and 10 white, 6 red and 9 black balls. One ball is taken from each urn. What is the probability that both will be of the same colour?

Step 1: Read the compressed wording as two urns: Urn 1 has 3 white, 7 red and 15 black balls (25 in all); Urn 2 has 10 white, 6 red and 9 black balls (25 in all). This reading is confirmed by the printed answer.

Step 2: P(both white) = (3/25) times (10/25) = 30/625.

Step 3: P(both red) = (7/25) times (6/25) = 42/625.

Step 4: P(both black) = (15/25) times (9/25) = 135/625.

Step 5: Adding them: (30 + 42 + 135)/625 = 207/625.

Answer: 207/625.

6.33 The probability that a man will be alive in 25 years is 3/5 and that his wife will be alive is 2/3. Find the probability that (i) both will be alive, (ii) only the man will be alive, (iii) only the wife will be alive, (iv) at least one will be alive, (v) none will be alive, (vi) at most one will be alive.

Step 1: P(man alive) = 3/5, P(wife alive) = 2/3, P(man dead) = 2/5, P(wife dead) = 1/3.

Step 2: (i) P(both alive) = (3/5) times (2/3) = 6/15.

Step 3: (ii) P(only man alive) = (3/5) times (1/3) = 3/15.

Step 4: (iii) P(only wife alive) = (2/5) times (2/3) = 4/15.

Step 5: (iv) P(at least one alive) = 1 - P(none alive) = 1 - (2/5) times (1/3) = 13/15.

Step 6: (v) P(none alive) = (2/5) times (1/3) = 2/15.

Step 7: (vi) P(at most one alive) = 1 - P(both alive) = 1 - 6/15 = 9/15.

Answer: (i) 6/15, (ii) 3/15, (iii) 4/15, (iv) 13/15, (v) 2/15, (vi) 9/15.

6.34 Three cards are drawn at random from a pack of 52. Find the probability that they consist of a Jack, a queen and a king.

Step 1: Total selections of 3 cards from 52 = C(52, 3) = 22100.

Step 2: Favourable selections = choose 1 Jack from 4, 1 queen from 4 and 1 king from 4 = 4 times 4 times 4 = 64.

Step 3: Probability = 64/22100.

Answer: 64/22100. Note: the printed key shows 64/20100, but C(52, 3) = 22100, so the denominator 20100 is a misprint.

6.35 Two cards are drawn from a pack of 52. Find the probability that both are aces, if the first card is (i) replaced, (ii) not replaced.

Step 1: (i) With replacement: P = (4/52) times (4/52) = 1/169.

Step 2: (ii) Without replacement: P = (4/52) times (3/51) = 12/2652 = 1/221.

Answer: (i) 1/169, (ii) 1/221.

6.36 Urn A contains 5 red and 3 white balls; Urn B contains 2 red and 6 white balls. Find the probability that (i) one ball drawn from each urn gives balls of the same colour, (ii) two balls drawn from each urn give all 4 balls of the same colour.

Step 1: (i) P(both red) = (5/8) times (2/8) = 10/64; P(both white) = (3/8) times (6/8) = 18/64. Adding: 28/64 = 7/16.

Step 2: (ii) All 4 red: (C(5, 2)/C(8, 2)) times (C(2, 2)/C(8, 2)) = (10/28) times (1/28) = 10/784. All 4 white: (C(3, 2)/C(8, 2)) times (C(6, 2)/C(8, 2)) = (3/28) times (15/28) = 45/784.

Step 3: Adding: (10 + 45)/784 = 55/784.

Answer: (i) 7/16, (ii) 55/784.

6.37 Three Ghori missiles are fired independently with hit probabilities 0.4, 0.5 and 0.6. Find the probability that at least 2 hit the target.

Step 1: Exactly 2 hit: (0.4 times 0.5 times 0.4) + (0.4 times 0.5 times 0.6) + (0.6 times 0.5 times 0.6) = 0.08 + 0.12 + 0.18 = 0.38.

Step 2: All 3 hit: 0.4 times 0.5 times 0.6 = 0.12.

Step 3: P(at least 2 hit) = 0.38 + 0.12 = 0.50.

Answer: 0.5.

6.38 A number X is chosen at random from the integers between 1 and 14. Find the probability that (i) X is a single-digit number, (ii) X is a multiple of 5 or 6.

Step 1: Take the integers 1 to 14 inclusive, 14 numbers in all (this reading matches the printed answers).

Step 2: (i) Single-digit numbers are 1 to 9, that is 9 numbers, so P = 9/14.

Step 3: (ii) Multiples of 5 are 5 and 10; multiples of 6 are 6 and 12; that is 4 numbers, so P = 4/14 = 2/7.

Answer: (i) 9/14, (ii) 2/7.

6.39 Find the odds in favour of an event whose probability is 4/7.

Step 1: P(event) = 4/7, so P(not event) = 3/7.

Step 2: Odds in favour = P(event) divided by P(not event) = (4/7) divided by (3/7) = 4/3.

Answer: Odds 4 to 3 in favour.

6.40 The odds are 9 to 7 against A, aged 35, living till he is 65, and 3 to 2 against B, aged 45, living till he is 75. Find the probability that one of these persons will be alive 30 years hence.

Step 1: P(A alive) = 7/16 and P(A dead) = 9/16; P(B alive) = 2/5 and P(B dead) = 3/5. Each period is 30 years, so the two events refer to the same future date.

Step 2: P(at least one alive) = 1 - P(both dead) = 1 - (9/16) times (3/5) = 1 - 27/80 = 53/80.

Step 3: For reference, P(exactly one alive) = (7/16) times (3/5) + (9/16) times (2/5) = 39/80.

Answer: 53/80, read as at least one of them alive; exactly one alive is 39/80. The printed key gives 53/80, the at-least-one value.

6.41 A coin is selected at random from one of two purses: purse 1 has 2 silver and 4 copper coins, purse 2 has 4 silver and 3 copper coins. Find the probability that the coin is (i) silver, (ii) copper.

Step 1: Each purse is chosen with probability 1/2.

Step 2: (i) P(silver) = (1/2) times (2/6) + (1/2) times (4/7) = 1/6 + 2/7 = 19/42.

Step 3: (ii) P(copper) = (1/2) times (4/6) + (1/2) times (3/7) = 1/3 + 3/14 = 23/42.

Answer: (i) 19/42, (ii) 23/42.

6.42 There are three purses: the first has 1 sovereign and 3 shillings, the second has 2 sovereigns and 4 shillings, and the third has 3 sovereigns and 1 shilling. A coin is taken from a purse selected at random. Find the chance that it is a sovereign.

Step 1: Each purse is chosen with probability 1/3.

Step 2: P(sovereign) = (1/3) times (1/4) + (1/3) times (2/6) + (1/3) times (3/4) = 1/12 + 1/9 + 1/4 = 16/36 = 4/9.

Answer: 4/9.

6.43 The probabilities of grades A, B and C in an examination are 0.09, 0.15 and 0.53. Find the probability of getting a grade lower than C.

Step 1: The grades A, B and C cover 0.09 + 0.15 + 0.53 = 0.77 of all outcomes.

Step 2: P(grade lower than C) = 1 - 0.77 = 0.23.

Answer: 0.23.

6.44 Three coins are tossed. Find the probability of getting at least one head.

Step 1: P(no head) = P(all tails) = (1/2)^3 = 1/8.

Step 2: P(at least one head) = 1 - 1/8 = 7/8.

Answer: 7/8.

6.45 A and B play 12 chess games: 6 are won by A, 4 by B and 2 are ties. In a tournament of 3 games, find the probability that (i) A wins all 3, (ii) two games end in a tie, (iii) A and B win alternately, (iv) B wins at least 1 game.

Step 1: P(A wins a game) = 6/12 = 1/2, P(B wins) = 4/12 = 1/3, P(tie) = 2/12 = 1/6.

Step 2: (i) P(A wins all 3) = (1/2)^3 = 1/8.

Step 3: (ii) P(two ties) = C(3, 2) times (1/6)^2 times (5/6) = 3 times (1/36) times (5/6) = 15/216 = 5/72.

Step 4: (iii) Alternating wins are A-B-A or B-A-B: (1/2) times (1/3) times (1/2) + (1/3) times (1/2) times (1/3) = 1/12 + 1/18 = 5/36.

Step 5: (iv) P(B wins at least 1) = 1 - (2/3)^3 = 1 - 8/27 = 19/27.

Answer: (i) 1/8, (ii) 5/72, (iii) 5/36, (iv) 19/27.

6.46 If A and B are independent events with P(A) = 0.25 and P(B) = 0.40, find P(A and B).

Step 1: For independent events, P(A and B) = P(A) times P(B) = 0.25 times 0.40 = 0.10.

Answer: 0.1.

6.47 A balanced die is rolled twice. Find the probability that the first roll is a five and the second is a number less than 4.

Step 1: P(first is 5) = 1/6. P(second less than 4) = 3/6 = 1/2.

Step 2: P = (1/6) times (1/2) = 1/12.

Answer: 1/12.

6.48 Two cards are drawn from a pack of 52 without replacement. Find the probability that both are diamonds.

Step 1: P(first diamond) = 13/52; P(second diamond) = 12/51.

Step 2: P = (13/52) times (12/51) = 156/2652 = 1/17.

Answer: 1/17.

6.49 A, B and C take turns throwing a die for a prize to be given to whoever first throws a 6. If A throws first, compare their chances.

Step 1: In each round of three throws, P(A wins the round) = 1/6, P(B wins) = (5/6) times (1/6) = 5/36, P(C wins) = (5/6) times (5/6) times (1/6) = 25/216, and P(no one wins) = (5/6)^3 = 125/216.

Step 2: Summing over all rounds (a geometric series with ratio 125/216): P(A) = (1/6) divided by (1 - 125/216) = 36/91; P(B) = (5/36) divided by (91/216) = 30/91; P(C) = (25/216) divided by (91/216) = 25/91.

Answer: The chances of A, B and C are in the ratio 36 : 30 : 25.

6.50 The first bag contains 4 white and 3 black balls; the second bag contains 3 white and 5 black balls. One ball is taken from the first bag and placed unseen in the second bag. Find the probability that a ball now drawn from the second bag is black.

Step 1: If a white ball is transferred (probability 4/7), the second bag has 4 white and 5 black balls, so P(black) = 5/9.

Step 2: If a black ball is transferred (probability 3/7), the second bag has 3 white and 6 black balls, so P(black) = 6/9.

Step 3: Total P(black) = (4/7) times (5/9) + (3/7) times (6/9) = 20/63 + 18/63 = 38/63.

Answer: 38/63.

6.51 A bag contains 4 white and 5 black balls. Two balls are drawn. Find the probability that they are of the same colour.

Step 1: Total selections of 2 balls from 9 = C(9, 2) = 36.

Step 2: Favourable selections = C(4, 2) + C(5, 2) = 6 + 10 = 16.

Step 3: Probability = 16/36.

Answer: 16/36.

6.52 There are three groups: the first has 3 girls and 1 boy, the second has 2 girls and 2 boys, and the third has 1 girl and 3 boys. One child is selected from each group. Find the chance that the selected children comprise one girl and two boys.

Step 1: The three orders giving one girl and two boys are: girl from group 1, boy from group 2, boy from group 3; boy, girl, boy; boy, boy, girl.

Step 2: P = (3/4) times (2/4) times (3/4) + (1/4) times (2/4) times (3/4) + (1/4) times (2/4) times (1/4) = 18/64 + 6/64 + 2/64 = 26/64 = 13/32.

Answer: 13/32.

6.53 A hits the target in 4 out of 5 shots, B in 2 out of 5 shots and C in 2 out of 4 shots. Taking one shot each, find the probability that (i) exactly 2 shots hit, (ii) at least two shots hit.

Step 1: P(A hits) = 4/5, P(B hits) = 2/5, P(C hits) = 1/2; the miss probabilities are 1/5, 3/5 and 1/2.

Step 2: (i) Exactly 2 hit: A and B hit, C misses: (4/5) times (2/5) times (1/2) = 8/50; A and C hit, B misses: (4/5) times (3/5) times (1/2) = 12/50; B and C hit, A misses:

(1/5) times (2/5) times (1/2) = 2/50. Total = 22/50 = 11/25.

Step 3: (ii) All 3 hit: (4/5) times (2/5) times (1/2) = 8/50. At least 2 hit = 22/50 + 8/50 = 30/50 = 60/100.

Answer: (i) 11/25, (ii) 60/100.

Exercise 6 - 6.54 Fill in the Blanks

6.54 (i) A set is any _____ collection of _____ things.

A set is any well defined collection of distinct things.

6.54 (ii) A set with only one element is singleton or unit set.

6.54 (iii) The set of subsets of a set is called power of the set.

6.54 (iv) Compound event contains more than one element.

6.54 (v) Two events are mutually exclusive if they have no common point.

6.54 (vi) If A union B = S, then A and B are collectively exhaustive events.

6.54 (vii) If n(A) = n(B), then A and B are equally likely events.

6.54 (viii) The number of subsets of a set containing n points is 2^n .

6.54 (ix) Orderly arrangements of r distinct things out of n are called _____ and denoted by _____.

Orderly arrangements of r distinct things out of n are called permutations and are denoted by nPr.

6.54 (x) A non-orderly combination arrangement is called combination.

Exercise 6 - 6.55 True or False

6.55 (i) The null set is also named the impossible event.

True

6.55 (ii) A part of a set is called an improper subset.

False

6.55 (iii) A subset of the sample space is called a sure event.

False

6.55 (iv) An event with only one sample point is called a compound event.

False

6.55 (v) If a coin is tossed four times, the number of sample points will be 22.

False

6.55 (vi) If A and B are mutually exclusive events, then P(A union B) = P(A) + P(B).

True

6.55 (vii) The probability of a red card from a pack of 52 cards is 13/15.

False

6.55 (viii) Complementary events are always mutually exclusive.

True

6.55 (ix) When a coin is tossed, the sample space is {HH, HT}.

False

6.55 (x) If A and B are independent events, then P(A union B) = P(A) + P(B).

False

Exercise 6 - Note on the Answers key

Note on the book's printed Answers key (p. 256): the key marks 6.55 (x) True.

The statement in 6.55 (x) says that if A and B are independent events, then P(A union B) = P(A) + P(B). For independent events the correct formula is P(A union B) = P(A) + P(B) - P(A)P(B); the addition formula without the product term holds for mutually exclusive events, not for independent events. The answer given here is therefore False, and the key's True is treated as a misprint.

Note on the 6.54 fills key (p. 256): the key prints an extra word 'Exhaustive'

between 6.54 (vi) and (viii).

The ten blanks of 6.54 are filled here with the standard words in order: (i) well defined, distinct; (ii) singleton or unit; (iii) power; (iv) compound; (v) mutually exclusive; (vi) collectively exhaustive; (vii) equally likely; (viii) 2^n; (ix) permutation, nPr; (x) combination. The key's extra word 'Exhaustive' is a duplicate of the word in (vi) and is not used as a separate blank.

Chapter 7: Random Variables

Exercise 7 - Questions 7.1 to 7.10

7.1 (i) Define a random variable and give an example. (ii) What are random numbers? How are they generated? Give an example of their application. (iii) Classify each of the following as a discrete or a continuous random variable: (a) the number of pages in a book, (b) the number of questions asked in an oral examination, (c) the life time of a light bulb, (d) the amount of rainfall at a particular location during different months of 1996.

(i) A random variable is a variable whose value is determined by the outcome of a random experiment. For example, if two coins are tossed, the number of heads obtained is a random variable, taking the values 0, 1 or 2 according to the outcome.

(ii) Random numbers are numbers produced by a random process in such a way that each digit has an equal probability of occurring. They are generated by a random device (such as a lottery machine) or by a pseudo-random formula, and they are read from a table of random numbers. They are used to draw random samples and to simulate random experiments without actually performing them, for example to simulate the tossing of a coin.

(iii) (a) Discrete random variable: the number of pages is counted in whole

numbers. (b) Discrete random variable: the number of questions is counted in whole numbers. (c) Continuous random variable: the life time can take any value in an interval of time. (d) Continuous random variable: the amount of rainfall can take any value in an interval.

7.2 Generate the first 6 random numbers using a pseudo-random generator with m = 100, a = 21, b = 7 and x0 = 10, using the recurrence x(n+1) = (a times x(n) + b) mod m.

Step 1: The recurrence is x(n+1) = (21 times x(n) + 7) mod 100, starting from x0 = 10.

Step 2: x1 = (21 times 10 + 7) mod 100 = 217 mod 100 = 17.

Step 3: x2 = (21 times 17 + 7) mod 100 = 364 mod 100 = 64.

Step 4: x3 = (21 times 64 + 7) mod 100 = 1351 mod 100 = 51.

Step 5: x4 = (21 times 51 + 7) mod 100 = 1078 mod 100 = 78.

Step 6: x5 = (21 times 78 + 7) mod 100 = 1645 mod 100 = 45.

Step 7: x6 = (21 times 45 + 7) mod 100 = 952 mod 100 = 52.

Answer: The first 6 generated numbers are 17, 64, 51, 78, 45 and 52.

7.3 Count the number of heads and tails when a single coin is tossed 10 times, without throwing a coin (that is, by simulation with random numbers).

Use one-digit random numbers from a random-number table. Take the digits 8, 3, 2,

6, 9, 8, 2, 8, 1, 2. Count an even digit as a head (H) and an odd digit as a tail (T).

The ten digits give, in order: H, T, H, H, T, H, H, H, T, H.

Counting them: the number of heads is 7 and the number of tails is 3. This is a specimen simulation: another set of random digits would give a different sequence, but the method is the same.

7.4 Two coins are tossed, and Y is the number of heads, with the distribution Y = 0 with probability 0.25, Y = 1 with probability 0.50 and Y = 2 with probability 0.25. Find a sequence of the number of heads in 20 trials, without throwing coins.

The sample space is S = {TT, HT, TH, HH}, and Y takes the values 0, 1 and 2 with probabilities 0.25, 0.50 and 0.25.

Simulate with two-digit random numbers: a number below 25 gives Y = 0, a number from 25 to 74 gives Y = 1, and a number from 75 upwards gives Y = 2. Using the generator of Question 7.2 continued (17, 64, 51, 78, 45, 52, 99, 86, 13, 80, 87, 34, 21, 48, 15, 22, 69, 56, 83, 50), the specimen sequence of 20 values is:

0, 1, 1, 2, 1, 1, 2, 2, 0, 2, 2, 1, 0, 1, 0, 0, 1, 1, 2, 1.

In this specimen, 0 occurs 5 times, 1 occurs 10 times and 2 occurs 5 times, close to the expected 5, 10 and 5. A student's own random numbers will give a different sequence; that is normal for a simulation.

7.5 Using the rule that the digits 0, 1, 2, 3 and 4 stand for a head and the digits 5, 6, 7, 8 and 9 stand for a tail, use random numbers to simulate 20 flips of a coin.

Take the digits of the generator of Question 7.2 in order. The first 20 digits are 1, 7, 6, 4, 5, 1, 7, 8, 4, 5, 5, 2, 9, 9, 8, 6, 1, 3, 8, 0.

Applying the rule (0 to 4 = head, 5 to 9 = tail) gives, in order: H, T, T, H, T, H, T, T, H, T, T, H, T, T, T, T, H, H, T, H.

Counting them: 8 heads and 12 tails in the 20 simulated flips. This is a specimen simulation; other random digits give a different sequence.

7.6 Two students choose either Mathematics (M) or Biology (B). If Y is the number of

students taking Biology, write down the sample space and the values of Y.

Each student chooses M or B, so the sample space is S = {BB, BM, MB, MM}.

Counting the number taking Biology in each outcome: BB gives 2, BM gives 1, MB gives 1 and MM gives 0.

So Y takes the values 0, 1 and 2.

7.7 Two coins are tossed and y is the number of heads obtained. Write down the sample space and the values of y.

The sample space is S = {TT, HT, TH, HH}.

Counting the heads in each outcome: TT gives 0, HT gives 1, TH gives 1 and HH gives 2.

So y takes the values 0, 1 and 2.

7.8 Three children are born in a family, each child equally likely to be a boy or a girl. If the random variable is the number of boys, write down all outcomes and the value of the variable for each outcome.

Writing B for a boy and G for a girl, the outcomes and the number of boys are: GGG gives 0; BGG gives 1; GBG gives 1; GGB gives 1; BBG gives 2; BGB gives 2; GBB gives 2; BBB gives 3.

So the random variable takes the values 0, 1, 2 and 3.

7.9 Four balls are drawn from a bag containing 5 white and 3 black balls. If X is the number of white balls drawn, find the possible values of X and the probability of each value.

Step 1: Four balls are drawn from 8 balls, so the total number of selections is C(8, 4) = 70.

Step 2: At most 3 black balls can be drawn, so at least 1 white ball is drawn; at most 4 white balls can be drawn. Hence X = 1, 2, 3, 4.

Step 3: P(X = 1) = C(5, 1) times C(3, 3) divided by 70 = 5 times 1 divided by 70 = 5/70.

Step 4: P(X = 2) = C(5, 2) times C(3, 2) divided by 70 = 10 times 3 divided by 70 = 30/70.

Step 5: P(X = 3) = C(5, 3) times C(3, 1) divided by 70 = 10 times 3 divided by 70 = 30/70.

Step 6: P(X = 4) = C(5, 4) times C(3, 0) divided by 70 = 5 times 1 divided by 70 = 5/70.

Answer: X = 1, 2, 3, 4 with probabilities 5/70, 30/70, 30/70 and 5/70. The probabilities add to 70/70 = 1.

7.10 Define a discrete random variable and a continuous random variable, giving an example of each.

A discrete random variable is a random variable that assumes only a finite (or countable) number of values, each value separated from the next. Example: the number of heads in 4 tosses of a coin, taking the values 0, 1, 2, 3 and 4.

A continuous random variable is a random variable that can assume every value in an interval. Example: the life time of a light bulb, which can be any positive length of time, or the height of a student, which can take any value in a range.

Exercise 7 - 7.11 Fill in the Blanks

7.11 (i) Random numbers are _____ by some _____ process.

Random numbers are generated by some random process.

7.11 (ii) Random numbers are obtained so that each digit has equal probability.

7.11 (iii) A random variable is also called chance or stochastic variable.

7.11 (iv) A random variable assuming only a finite number of values is discrete random variable.

7.11 (v) A random variable assuming all possible values in a _____ is _____ random variable.

A random variable assuming all possible values in a range is a continuous random variable.

7.11 (vi) The sum of the probabilities of the events of a sample space is always 1 .

7.11 (vii) A probability function is a mathematical function.

7.11 (viii) The mean of a probability distribution is called mathematical expectation .

7.11 (ix) E(X) = sum X f(X) if it converges absolutely.

7.11 (x) A random variable may be _____ or _____.

A random variable may be discrete or continuous.

Chapter 8: Probability Distributions

Exercise 8 - Questions 8.1 to 8.37

8.1 What are random numbers? How are they generated? Give some applications of random numbers.

Random numbers are numbers produced by a random process so that every digit has the same chance of appearing. They are generated by a random device, such as a lottery machine, or by a formula like a pseudo-random generator, and they are collected in tables of random numbers.

They are used to select random samples, to simulate experiments that are costly or

impossible to repeat, and to assign treatments at random in experiments.

8.2 Three balls are drawn from a bag containing 5 white and 3 black balls. If X is the number of white balls drawn, find the probability distribution of X.

Step 1: Total selections of 3 balls from 8 = C(8, 3) = 56.

Step 2: P(X = 0) = C(5, 0) times C(3, 3) divided by 56 = 1/56.

Step 3: P(X = 1) = C(5, 1) times C(3, 2) divided by 56 = 15/56.

Step 4: P(X = 2) = C(5, 2) times C(3, 1) divided by 56 = 30/56.

Step 5: P(X = 3) = C(5, 3) times C(3, 0) divided by 56 = 10/56.

Answer: X = 0, 1, 2, 3 with probabilities 1/56, 15/56, 30/56, 10/56.

8.3 Seven candidates apply for three typist positions. Four of them know Urdu typing and three do not. Find the probability distribution of the number of selected candidates who know Urdu typing.

Step 1: Total selections of 3 candidates from 7 = C(7, 3) = 35.

Step 2: P(X = 0) = C(4, 0) times C(3, 3) divided by 35 = 1/35.

Step 3: P(X = 1) = C(4, 1) times C(3, 2) divided by 35 = 12/35.

Step 4: P(X = 2) = C(4, 2) times C(3, 1) divided by 35 = 18/35.

Step 5: P(X = 3) = C(4, 3) times C(3, 0) divided by 35 = 4/35.

Answer: X = 0, 1, 2, 3 with probabilities 1/35, 12/35, 18/35, 4/35.

8.4 (i) Define a probability distribution, and distinguish between a discrete and a continuous probability distribution, with examples. (ii) A coin is tossed 4 times. If X is the number of tails, find the probability distribution of X and draw its probability histogram. (iii) A bag contains 4 red and 6 black balls. A sample of 4 balls is drawn without replacement. If X is the number of red balls, find the probability distribution of X.

(i) A probability distribution is the list of all possible values of a random variable together with their probabilities. A discrete probability distribution gives the probabilities of separate, countable values, for example the number of heads in 3 tosses. A continuous probability distribution describes a variable that takes values in an interval, through a density function, for example the waiting time for a bus.

(ii) A coin is tossed 4 times and X is the number of tails. X = 0, 1, 2, 3, 4 with probabilities 1/16, 4/16, 6/16, 4/16, 1/16. Probability histogram: draw rectangles of width 1 centred at x = 0, 1, 2, 3, 4 with heights 1/16, 4/16, 6/16, 4/16 and 1/16.

(iii) A sample of 4 is drawn without replacement from 4 red and 6 black balls, and X is the number of red balls. Total selections = C(10, 4) = 210. X = 0, 1, 2, 3, 4 with probabilities 15/210, 80/210, 90/210, 24/210 and 1/210.

8.5 (i) Define a probability function and give an example. (ii) Two fair dice are thrown and Y is the product of the two scores. Find the probability distribution of Y.

(i) A probability function f(x) assigns to each value x of a discrete random variable its probability P(X = x). Example: for one throw of a die, f(x) = 1/6 for x = 1, 2, 3, 4, 5, 6.

(ii) Let Y be the product of the scores on two fair dice. Counting the 36 equally likely pairs gives: Y = 1 with probability 1/36; 2 with 2/36; 3 with 2/36; 4 with 3/36; 5 with 2/36; 6 with 4/36; 8 with 2/36; 9 with 1/36; 10 with 2/36; 12 with 4/36; 15 with 2/36;

16 with 1/36; 18 with 2/36; 20 with 2/36; 24 with 2/36; 25 with 1/36; 30 with 2/36;

36 with 1/36. The probabilities add to 36/36 = 1.

8.6 (i) State the properties that a function must satisfy to be a probability function or a probability density function. (ii) Which of the following are probability functions? (a) f(y) = 1/4 for y = 1, 2, 3, 4, 5; (b) f(y) = 1/y for y = 1, 2, 3, 4; (c) f(y) = y/15 for y = 0, 1, 2, 3, 4, 5; (d) f(y) = (5 - y^2)/6 for y = 0, 1, 2.

(i) For a discrete probability function: every value f(x) is at least 0, and the sum of f(x) over all values is 1. For a probability density function: f(x) is at least 0 for every x, and the total area under the curve is 1.

(ii) (a) f(y) = 1/4 for y = 1, 2, 3, 4, 5: the sum is 5/4, not 1, so No. (b) f(y) = 1/y for y = 1, 2, 3, 4: the sum is 1 + 1/2 + 1/3 + 1/4 = 25/12, not 1, so No. (c) f(y) = y/15 for y = 0, 1, 2, 3, 4, 5: the sum is (0 + 1 + 2 + 3 + 4 + 5)/15 = 1 and every value is at least 0, so Yes. (d) f(y) = (5 - y^2)/6 for y = 0, 1, 2: the sum is (5 + 4 + 1)/6 = 10/6, not 1, so No.

8.7 Find the value of c so that each of the following is a probability function: (a) f(y) = c times y for Y = 1, 2, 3, 4, 5; (b) f(y) = (1 - c) times c^y for Y = 0, 1, 2, ...

Step 1: (a) The sum over y = 1 to 5 of c times y is c times (1 + 2 + 3 + 4 + 5) = 15c. Setting 15c = 1 gives c = 1/15.

Step 2: (b) The sum over y = 0, 1, 2, ... of (1 - c) times c^y is (1 - c) times (1 + c + c^2 + ...) = (1 - c) divided by (1 - c) = 1, provided the geometric series converges, that is 0 < c < 1. (The value c = 0 gives a degenerate distribution concentrated at y = 0 and is excluded.)

Answer: (a) c = 1/15, (b) 0 < c < 1.

8.8 A random variable Y takes the values 0, 1, 2 and 3 with probabilities (1 + 3

times theta)/4, (1 - theta)/4, (1 + 2 times theta)/4 and (1 - 4 times theta)/4. For what values of theta is this a valid probability distribution?

Step 1: The four probabilities add to (1 + 3 theta + 1 - theta + 1 + 2 theta + 1 - 4 theta)/4 = 4/4 = 1 for every theta, so only the non-negativity conditions matter.

Step 2: (1 + 3 theta)/4 at least 0 gives theta at least -1/3. (1 - theta)/4 at least 0 gives theta at most 1. (1 + 2 theta)/4 at least 0 gives theta at least -1/2. (1 - 4 theta)/4 at least 0 gives theta at most 1/4.

Step 3: Combining them: -1/3 at most theta at most 1/4. The printed key writes this with strict inequalities, -1/3 < theta < 1/4; at the end points one probability is zero.

Answer: -1/3 < theta < 1/4 (end points allowed if a zero probability is accepted).

8.9 (i) For the distribution x = 0, 1, 2, 3, 4 with P(x) = 1/126, 20/126, 60/126, 40/126, 5/126, verify that E(2X + 3) = 2 E(X) + 3. (ii) For the distribution x = -1, 0, 1, 2, 3 with P(x) = 0.125, 0.500, 0.200, 0.050, 0.125, find (a) E(X) and Var(X), (b) the distribution of Y = 2X + 1 and then E(Y) and Var(Y) from the distribution of Y.

Step 1: (i) E(X) = (0 times 1 + 1 times 20 + 2 times 60 + 3 times 40 + 4 times 5)/126 = 280/126 = 2.2222. Then E(2X + 3) = 2 times 2.2222 + 3 = 7.4444, about 7.44, and 2 E(X) + 3 is the same, so the rule is verified.

Step 2: (ii)(a) E(X) = (-1) times 0.125 + 0 times 0.500 + 1 times 0.200 + 2 times 0.050 + 3 times 0.125 = 0.55. E(X^2) = 1 times 0.125 + 0 + 1 times 0.200 + 4 times 0.050 + 9 times 0.125 = 1.65. Var(X) = 1.65 - (0.55)^2 = 1.3475.

Step 3: (ii)(b) Y = 2X + 1 takes the values -1, 1, 3, 5, 7 with the same probabilities

0.125, 0.500, 0.200, 0.050, 0.125. E(Y) = (-1) times 0.125 + 1 times 0.500 + 3 times 0.200 + 5 times 0.050 + 7 times 0.125 = 2.1, and Var(Y) = 4 times Var(X) = 5.39.

Answer: (i) E(2X + 3) = 7.44 = 2 E(X) + 3; (ii)(a) E(X) = 0.55, Var(X) = 1.3475; (ii)(b) E(Y) = 2.1, Var(Y) = 5.39.

8.10 The distribution of Y, the number of courses taught by a teacher in the spring semester, is: y = 1, 2, 3, 4, 5 with P(y) = 0.1, 0.2, 0.3, 0.3, 0.1. Find (i) P(a teacher taught 2 courses), (ii) P(less than 4 courses), (iii) P(between 2 and 5 courses), (iv) P(at least 3 courses), (v) P(at most 4 courses).

Step 1: (i) P(Y = 2) = 0.2.

Step 2: (ii) P(Y less than 4) = 0.1 + 0.2 + 0.3 = 0.6.

Step 3: (iii) Taking 'between 2 and 5' strictly, P(2 < Y < 5) = P(3) + P(4) = 0.3 + 0.3 = 0.6.

Step 4: (iv) P(Y at least 3) = 0.3 + 0.3 + 0.1 = 0.7.

Step 5: (v) P(Y at most 4) = 0.1 + 0.2 + 0.3 + 0.3 = 0.9.

Answer: (i) 0.2, (ii) 0.6, (iii) 0.6, (iv) 0.7, (v) 0.9.

8.11 A box contains five slips numbered 1, 2, 3, 4, 5. Two slips are selected without replacement. Find the possible values of (i) the sum of the two numbers, (ii) the difference between the first and the second number.

(i) The two numbers are chosen from 1, 2, 3, 4, 5 without replacement. The smallest sum is 1 + 2 = 3 and the largest is 4 + 5 = 9, and every value in between occurs. So the possible sums are {3, 4, 5, 6, 7, 8, 9}.

(ii) The difference between the first and the second number, taken as an absolute difference, can be 1, 2, 3 or 4. So the possible values are {1, 2, 3, 4}. With signs, the differences run from -4 to 4 excluding 0.

8.12 Four students are asked whether they are satisfied with a service, answering satisfactory (S) or not satisfactory (NS). If Y is the number answering satisfactory, write down the possible outcomes and the values of Y.

The four students can each answer satisfactory (S) or not satisfactory (NS). The outcomes and the number saying satisfactory are: NNNN gives 0; SNNN, NSNN, NNSN and NNNS give 1; SSNN, SNSN, SNNS, NSSN, NSNS and NNSS give 2; SSSN, SSNS, SNSS and NSSS give 3; SSSS gives 4.

So Y takes the values 0, 1, 2, 3 and 4.

8.13 A point is selected at random on the surface of a lake whose maximum depth is 30 feet. If y is the depth at the selected point, what are the possible values of y? Is y discrete or continuous?

The depth at a point chosen at random on the lake can be any value from the surface down to the maximum depth, so y takes every value between 0 and 30 feet. Since y can take any value in an interval, it is a continuous random variable.

8.14 Find the mean and variance of the distribution y = 0, 1, 2, 3, 4, 5 with P(y) = 0.1, 0.2, 0.3, 0.2, 0.1, 0.1.

Step 1: Mean = 0 times 0.1 + 1 times 0.2 + 2 times 0.3 + 3 times 0.2 + 4 times 0.1 + 5 times 0.1 = 2.3.

Step 2: E(Y^2) = 0 + 1 times 0.2 + 4 times 0.3 + 9 times 0.2 + 16 times 0.1 + 25 times 0.1 = 7.3.

Step 3: Variance = 7.3 - (2.3)^2 = 7.3 - 5.29 = 2.01.

Answer: Mean = 2.3, Variance = 2.01.

8.15 (i) A continuous random variable X takes values from 0 to 4 with density f(x) = 1/2 - a times x. Find (a) a, (b) P(1 < X < 2). (ii) A density is f(x) = c times x for 0 < X < 2. Find (a) c, (b) P(1 < X < 1.5), (c) P(X < 1.5). (iii) A density is f(x) = k times x^2 for 0 < X < 1. Determine k and find P(1/3 < X < 1/2).

Step 1: (i)(a) The total area must be 1: the integral from 0 to 4 of (1/2 - a x) dx = 2 - 8a = 1, so a = 1/8.

Step 2: (i)(b) P(1 < X < 2) = the integral from 1 to 2 of (1/2 - x/8) dx = 0.75 - 0.4375 = 0.3125 = 5/16.

Step 3: (ii)(a) The integral from 0 to 2 of c x dx = 2c = 1, so c = 1/2.

Step 4: (ii)(b) P(1 < X < 1.5) = the integral from 1 to 1.5 of (x/2) dx = (1/4) times (1.5^2 - 1^2) = 0.3125.

Step 5: (ii)(c) P(X < 1.5) = the integral from 0 to 1.5 of (x/2) dx = (1/4) times 1.5^2 = 0.5625.

Step 6: (iii) The integral from 0 to 1 of k x^2 dx = k/3 = 1, so k = 3, and f is a probability density function for a continuous variable. P(1/3 < X < 1/2) = the integral from 1/3 to 1/2 of 3x^2 dx = (1/2)^3 - (1/3)^3 = 1/8 - 1/27 = 19/216.

Answer: (i) a = 1/8, P = 5/16; (ii) c = 1/2, P = 0.3125, P = 0.5625; (iii) k = 3, P =

19/216.

8.16 Define a random variable, and distinguish between a discrete and a continuous random variable, with examples.

A random variable is a variable whose value depends on the outcome of a random experiment. A discrete random variable takes only separate, countable values, for example the number of heads in 5 tosses. A continuous random variable takes every value in an interval, for example the height of a student or the time taken to complete a task.

8.17 Find the probability distribution of the number of boys in families with three children, assuming boys and girls are equally likely.

Step 1: With n = 3 and p = 1/2, P(X = x) = C(3, x) divided by 8.

Step 2: X = 0, 1, 2, 3 with probabilities 1/8, 3/8, 3/8, 1/8.

Answer: 1/8, 3/8, 3/8, 1/8.

8.18 A lot of 12 items contains 4 defective items. Five items are chosen at random. If X is the number of defective items, find (i) the probability distribution of X, (ii) P(X at most 1), (iii) verify that the sum of all P(x) is 1.

Step 1: Total selections = C(12, 5) = 792. P(X = x) = C(4, x) times C(8, 5-x) divided by 792.

Step 2: (i) X = 0, 1, 2, 3, 4 with probabilities 56/792, 280/792, 336/792, 112/792, 8/792.

Step 3: (ii) P(X at most 1) = (56 + 280)/792 = 336/792.

Step 4: (iii) The sum is (56 + 280 + 336 + 112 + 8)/792 = 792/792 = 1, as required.

Answer: (i) 56/792, 280/792, 336/792, 112/792, 8/792; (ii) 336/792; (iii) verified.

8.19 (i) Define a continuous random variable and its probability distribution. (ii) Find the constant k so that f(x) = 1/k for a at most X at most b, and 0 elsewhere, is a density function.

(i) A continuous random variable is one that can take every value in an interval. Its probability distribution is given by a density function f(x): the probability that X lies between a and b is the area under the curve of f between a and b.

(ii) The total area must be 1: the integral from a to b of (1/k) dx = (b - a)/k = 1, so k = b - a.

8.20 (i) Define expected value and state the properties of expectation. (ii) For the distribution x = 0, 1, 2, 3, 4, 5 with P(x) = 6/36, 10/36, 8/36, 6/36, 4/36, 2/36, find the mean, variance, standard deviation and coefficient of variation.

Step 1: (i) The expected value of X is E(X) = sum of x times P(x) over all values. Its properties are: E(a constant) = the constant; E(aX) = a E(X); E(X + Y) = E(X) + E(Y); and for a constant a, E(X + a) = E(X) + a. Also Var(aX + b) = a^2 Var(X).

Step 2: (ii) Mean = (0 times 6 + 1 times 10 + 2 times 8 + 3 times 6 + 4 times 4 + 5 times 2)/36 = 70/36 = 1.94.

Step 3: E(X^2) = (0 + 10 + 32 + 54 + 64 + 50)/36 = 210/36 = 5.8333. Variance = 5.8333 - (1.94)^2 = 2.05, so S.D. = 1.43.

Step 4: Coefficient of variation = (1.43/1.94) times 100 = 73.68 percent (the printed key gives 73.71 percent, a rounding difference).

Answer: Mean = 1.94, S.D. = 1.43, C.V. = 73.68 percent.

8.21 For the distribution x = 1, 2, 3, 4, 5 with f(x) = 0.125, 0.45, 0.25, 0.05, 0.125, find the mean and variance.

Step 1: Mean = 1 times 0.125 + 2 times 0.45 + 3 times 0.25 + 4 times 0.05 + 5 times 0.125 = 2.6.

Step 2: E(X^2) = 1 times 0.125 + 4 times 0.45 + 9 times 0.25 + 16 times 0.05 + 25 times 0.125 = 8.1.

Step 3: Variance = 8.1 - (2.6)^2 = 8.1 - 6.76 = 1.34.

Answer: Mean = 2.6, Variance = 1.34.

8.22 (i) A density is f(x) = (x + 1)/8 for X from 2 to 4. Find (a) P(X < 3.5), (b) P(2.4 < X < 3.5), (c) P(X = 1.5). (ii) A density is f(x) = 2x for 0 at most x at most 1. Find (a) P(X = 1/2), (b) P(X > 1/4), (c) P(1/4 < X < 1/2).

Step 1: (i) The distribution function is F(x) = (x^2/16 + x/8) - (4/16 + 2/8) for x between 2 and 4, that is F(x) = x^2/16 + x/8 - 0.5.

Step 2: (i)(a) P(X < 3.5) = F(3.5) = 0.765625 + 0.4375 - 0.5 = 0.7031.

Step 3: (i)(b) P(2.4 < X < 3.5) = F(3.5) - F(2.4) = 1.203125 - 0.66 = 0.543125.

Step 4: (i)(c) P(X = 1.5) = 0, both because X is continuous and because 1.5 is outside the range 2 to 4.

Step 5: (ii)(a) For a continuous variable, P(X = 1/2) = 0. The value 1/4 printed in the key is P(X < 1/2) = the integral from 0 to 1/2 of 2x dx = 1/4.

Step 6: (ii)(b) P(X > 1/4) = 1 - the integral from 0 to 1/4 of 2x dx = 1 - 1/16 = 15/16.

Step 7: (ii)(c) P(1/4 < X < 1/2) = 1/4 - 1/16 = 3/16.

Answer: (i) 0.7031, 0.543125, 0; (ii) 0 (with P(X < 1/2) = 1/4), 15/16, 3/16.

8.23 A continuous random variable X takes values between 2 and 8 inclusive, with density a times (x + 30). Find (i) a, (ii) P(3 < X < 6), (iii) P(X at most 6), (iv) P(X at least 4).

Step 1: (i) The total area is a times the integral from 2 to 8 of (x + 30) dx = a times (32 + 240 - 2 - 60) = 210a = 1, so a = 1/210.

Step 2: (ii) P(3 < X < 6) = (1/210) times (18 + 180 - 4.5 - 90) = 103.5/210 = 0.4928.

Step 3: (iii) P(X at most 6) = (1/210) times (18 + 180 - 2 - 60) = 136/210 = 0.6476.

Step 4: (iv) P(X at least 4) = (1/210) times (32 + 240 - 8 - 120) = 144/210 = 0.6857 (the printed key gives 0.6851, a rounding difference).

Answer: (i) a = 1/210, (ii) 0.4928, (iii) 0.6476, (iv) 0.6857.

8.24 A dealer earns Rs. 800 per day if the weather is hot, Rs. 200 per day if it is fair, and loses Rs. 50 per day if it is cloudy. If P(hot) = 0.40 and P(cloudy) = 0.35, find

the dealer's expected daily earning.

Step 1: P(fair) = 1 - 0.40 - 0.35 = 0.25.

Step 2: Expected earning = 800 times 0.40 + 200 times 0.25 - 50 times 0.35 = 320 + 50 - 17.5 = 352.5.

Answer: Rs. 352.5.

8.25 A committee of 5 is chosen from 3 females and 5 males. Find the expected number of female members.

The number of female members in a committee of 5 chosen from 3 females and 5 males is hypergeometric with mean n times K divided by N = 5 times 3 divided by 8 = 15/8 = 1.875.

8.26 State the properties of a probability function and of a probability density function.

For a discrete probability function: f(x) is at least 0 for every x, and the sum of f(x) over all x is 1. For a probability density function: f(x) is at least 0 for every x, and the total area under the curve of f is 1. In both cases probabilities lie between 0 and 1, and the probability of the whole sample space is 1.

8.27 (i) Explain the concept of a distribution function. (ii) Ten vegetable cans, 5 of tomatoes and 5 of corns, are placed in a box and 5 are selected. Find the probability distribution and the distribution function of the number of tomato cans selected.

Step 1: (i) The distribution function F(x) = P(X at most x) gives, for each x, the total probability accumulated up to x. It never decreases, it is 0 before the smallest value and 1 from the largest value onwards.

Step 2: (ii) Total selections = C(10, 5) = 252. The probability function is: P(0) =

1/252, P(1) = 25/252, P(2) = 100/252, P(3) = 100/252, P(4) = 25/252, P(5) = 1/252.

Step 3: Adding them up gives the distribution function: F(0) = 1/252, F(1) = 26/252, F(2) = 126/252, F(3) = 226/252, F(4) = 251/252, F(5) = 252/252.

Answer: Probability function 1/252, 25/252, 100/252, 100/252, 25/252, 1/252; distribution function 1/252, 26/252, 126/252, 226/252, 251/252, 252/252.

8.28 The number of automobile accidents in a city takes the values 0, 1, 2, 3 with probabilities 1/8, 2/8, 2/8, 3/8. Find the expected value (mean) of the number of accidents.

Step 1: Mean = 0 times 1/8 + 1 times 2/8 + 2 times 2/8 + 3 times 3/8 = (0 + 2 + 4 + 9)/8 = 15/8 = 1.875.

Answer: 1.875. Note: the question as printed lists five values, '1, 2, 3, 4, 4', against only four probabilities; the key's mean 1.875 follows from the values 0, 1, 2, 3 with those probabilities, and the question is solved that way here.

8.29 A and B throw a die for a prize of Rs. 11. The winner is the first to throw a 6, and A throws first. Find the respective expectations of A and B.

Step 1: P(A wins) = (1/6) divided by (1 - (5/6)^2) = (1/6) divided by (11/36) = 6/11. P(B wins) = 5/11.

Step 2: A's expectation = 11 times 6/11 = Rs. 6.0. B's expectation = 11 times 5/11 = Rs. 5.0.

Answer: A Rs. 6.0, B Rs. 5.0.

8.30 A bag contains 2 white and 2 black balls. A, B and C draw balls one by one without replacement. Whoever draws a white ball first receives Rs. 12. Find the respective expectations of A, B and C.

Step 1: P(A draws white first) = 2/4 = 1/2, so A's expectation = 12 times 1/2 = Rs. 6.0.

Step 2: P(B draws white first) = (2/4) times (2/3) = 1/3, so B's expectation = 12 times 1/3 = Rs. 4.0.

Step 3: P(C draws white first) = (2/4) times (1/3) times (2/2) = 1/6, so C's expectation = 12 times 1/6 = Rs. 2.0.

Answer: A Rs. 6.0, B Rs. 4.0, C Rs. 2.0.

8.31 Three balls are drawn from a bag of 5 white and 3 black balls. If X is the number of white balls drawn, find the probability distribution of X, its mean and variance.

Step 1: The distribution is the same as in Question 8.2: X = 0, 1, 2, 3 with probabilities 1/56, 15/56, 30/56, 10/56.

Step 2: Mean = (0 times 1 + 1 times 15 + 2 times 30 + 3 times 10)/56 = 105/56 = 1.875.

Step 3: E(X^2) = (0 + 15 + 120 + 90)/56 = 225/56 = 4.0179. Variance = 4.0179 - (1.875)^2 = 0.5024.

Answer: Distribution 1/56, 15/56, 30/56, 10/56; Mean = 105/56; Variance = 0.5024.

8.32 A coin is biased so that a head is thrice as likely as a tail. Find the probability distribution of the number of heads, and its mean and variance, when the coin is tossed 4 times.

Step 1: P(head) = 3/4 and P(tail) = 1/4. With n = 4, P(X = x) = C(4, x) times (3/4)^x times (1/4)^(4-x).

Step 2: X = 0, 1, 2, 3, 4 with probabilities 1/256, 12/256, 54/256, 108/256, 81/256.

Step 3: Mean = np = 4 times 3/4 = 3. Variance = npq = 4 times 3/4 times 1/4 = 0.75.

Answer: Distribution 1/256, 12/256, 54/256, 108/256, 81/256; Mean = 3, Variance = 0.75.

8.33 About 10 percent of glass bottles have serious defects. If 2 bottles are selected, find the expected number of defective bottles.

Step 1: With n = 2 and p = 0.1, the distribution is: P(0) = 0.81, P(1) = 0.18, P(2) = 0.01. (The printed key shows 0.1 for x = 2, but the three probabilities must add to 1, so the value is 0.01.)

Step 2: Expected number = np = 2 times 0.1 = 0.2. Variance = npq = 0.18.

Answer: Distribution 0.81, 0.18, 0.01; Mean = 0.2, Variance = 0.18.

8.34 A random variable X takes the values -3, -2, 2, 3, 4 with probabilities 1/5, 1/10, 1/10, 1/5, 2/5. Compute E(X), show that E(5X + 10) = 5 E(X) + 10, compute Var(X) and Var(5X + 10), and find the ratio of the two variances.

Step 1: E(X) = (-3) times 1/5 + (-2) times 1/10 + 2 times 1/10 + 3 times 1/5 + 4 times 2/5 = -0.6 - 0.2 + 0.2 + 0.6 + 1.6 = 1.6 = 8/5.

Step 2: E(X^2) = 9 times 1/5 + 4 times 1/10 + 4 times 1/10 + 9 times 1/5 + 16 times 2/5 = 10.8. Var(X) = 10.8 - (1.6)^2 = 8.24.

Step 3: E(5X + 10) = 5 times 1.6 + 10 = 18, which equals 5 E(X) + 10, as required.

Step 4: Var(5X + 10) = 25 times Var(X) = 25 times 8.24 = 206.

Step 5: Ratio Var(X) : Var(5X + 10) = 8.24 : 206 = 1 : 25.

Answer: E(X) = 8/5, Var(X) = 8.24, E(5X + 10) = 18, Var(5X + 10) = 206, ratio 1 : 25.

8.35 A probability function is f(x) = (6 - |7 - x|)/36 for X = 2, 3, ..., 12. Find the mean and variance.

Step 1: The values of f(x) for x = 2 to 12 are 1/36, 2/36, 3/36, 4/36, 5/36, 6/36, 5/36, 4/36, 3/36, 2/36, 1/36, which add to 1.

Step 2: The distribution is symmetric about 7, so the mean is 7.

Step 3: E(X^2) = (4 times 1 + 9 times 2 + 16 times 3 + 25 times 4 + 36 times 5 + 49 times 6 + 64 times 5 + 81 times 4 + 100 times 3 + 121 times 2 + 144 times 1)/36 = 1974/36 = 54.8333.

Step 4: Variance = 54.8333 - 49 = 5.8333.

Answer: Mean = 7, Variance = 5.83.

8.36 For the probability distribution x = -10, -20, 30 with P(x) = 1/5, 3/10, 1/2, find (i) E(X), (ii) E(X^2), (iii) Var(X) = E[(X - E(X))^2].

Step 1: (i) E(X) = (-10) times 1/5 + (-20) times 3/10 + 30 times 1/2 = -2 - 6 + 15 = 7.

Step 2: (ii) E(X^2) = 100 times 1/5 + 400 times 3/10 + 900 times 1/2 = 20 + 120 + 450 = 590.

Step 3: (iii) Variance = 590 - 49 = 541.

Answer: (i) 7, (ii) 590, (iii) 541.

8.37 (i) Define expectation of a random variable. (ii) The probability distribution of a discrete random variable X is f(x) = C(3, x) times (1/4)^x times (3/4)^(3-x) for x = 0, 1, 2, 3. Find E(X) and E(X^2).

Step 1: (i) The expectation (mean) of X is E(X) = sum of x times f(x) over all values of x.

Step 2: (ii) This is a binomial distribution with n = 3 and p = 1/4, so E(X) = np = 3/4 = 0.75.

Step 3: Variance = npq = 3 times 1/4 times 3/4 = 0.5625, so E(X^2) = variance + (mean)^2 = 0.5625 + 0.5625 = 1.125.

Answer: E(X) = 0.75, E(X^2) = 1.125.

Exercise 8 - 8.38 True or False

8.38 (i) A random variable is also named as a chance variable.

True

8.38 (ii) The number of accidents occurring on G.T. road during one month is the example of a continuous random variable.

False

8.38 (iii) The probability cannot exceed 1.

True

8.38 (iv) The range of a continuous random variable is from '0' to 'n'.

False

8.38 (v) The distribution function is an increasing function.

True

8.38 (vi) The expectation of a random variable is also named as the mean of a random variable.

True

8.38 (vii) The probability function can be negative.

False

8.38 (viii) A discrete probability distribution is represented by an area graph.

False

8.38 (ix) If X and Y are independent random variables then $E(XY) = E(X)\,E(Y)$.

True

8.38 (x) If X and Y are independent random variables, then $S.D.(X - Y) = S.D.(X) - S.D.(Y)$.

False

Exercise 8 - Note on the Answers key

Note on the book's printed Answers key (p. 260): the key marks 8.38 (iii) F, (v) F, (viii) T, (ix) F, (x) T.

The answers given here follow the chapter's definitions, and differ from the key at five places. (iii) A probability cannot exceed 1, so the statement is True. (v) The distribution function is an increasing (non-decreasing) function, so the statement is True. (viii) A discrete probability distribution is shown by a probability histogram of bars, not by an area graph, so the statement is False. (ix) For independent random variables X and Y, E(XY) = E(X)E(Y), so the statement is True. (x) For independent X and Y, Var(X - Y) = Var(X) + Var(Y), so S.D.(X - Y) is the square root of the sum of the variances, not the difference of the standard deviations, and the statement is False. The five key entries are treated as misprints.

Note on 8.15 (iii): the printed key (p. 258) gives k = 2 and 13/216.

For the density in 8.15 (iii), the normalising condition forces k = 3, and then P(1/3 < X < 1/2) = 1/8 - 1/27 = 19/216. The key's k = 2 and 13/216 do not satisfy the normalising condition, and are treated as misprints.

Chapter 9: Binomial and Hypergeometric Probability Distribution

Exercise 9 - Questions 9.1 to 9.23

9.1 Define the binomial probability distribution.

A binomial probability distribution is the probability distribution of the number of successes in a fixed number of independent Bernoulli trials, each trial having the same probability p of success. If X is the number of successes in n trials, then P(X = x) = C(n, x) times p^x times q^(n-x), for x = 0, 1, 2, ..., n, where q = 1 - p. Its mean is np and its variance is npq.

9.2 Define a binomial experiment and state its properties.

A binomial experiment is an experiment with the following properties: (i) it consists of a fixed number n of trials; (ii) each trial has only two possible outcomes, success and failure; (iii) the probability of success p is the same in every trial; (iv) the trials are independent of one another.

For example, tossing a fair coin 5 times and counting the number of heads is a binomial experiment with n = 5 and p = 1/2.

9.3 If the probability of an event is p = 3/8, complete the binomial distribution for n = 5 trials.

Step 1: Here n = 5, p = 3/8 and q = 1 - 3/8 = 5/8. The formula is P(X = x) = C(5, x) times (3/8)^x times (5/8)^(5-x).

Step 2: P(X = 0) = (5/8)^5 = 3125/32768.

Step 3: P(X = 1) = 5 times (3/8) times (5/8)^4 = 9375/32768.

Step 4: P(X = 2) = 10 times (3/8)^2 times (5/8)^3 = 11250/32768.

Step 5: P(X = 3) = 10 times (3/8)^3 times (5/8)^2 = 6750/32768.

Step 6: P(X = 4) = 5 times (3/8)^4 times (5/8) = 2025/32768.

Step 7: P(X = 5) = (3/8)^5 = 243/32768.

Answer: x = 0, 1, 2, 3, 4, 5 with probabilities 3125/32768, 9375/32768, 11250/32768, 6750/32768, 2025/32768 and 243/32768.

9.4 A fair coin is tossed four times. Find the probability of (i) 4 heads, (ii) 1 tail and 3 heads, (iii) at least 2 heads, (iv) at most 2 heads.

Step 1: Here n = 4 and p = 1/2, so P(X = x) = C(4, x) divided by 16.

Step 2: (i) P(4 heads) = C(4, 4) divided by 16 = 1/16.

Step 3: (ii) 1 tail and 3 heads means X = 3, so P = C(4, 3) divided by 16 = 4/16 = 1/4.

Step 4: (iii) P(at least 2 heads) = P(2) + P(3) + P(4) = (6 + 4 + 1) divided by 16 = 11/16.

Step 5: (iv) P(at most 2 heads) = P(0) + P(1) + P(2) = (1 + 4 + 6) divided by 16 = 11/16.

Answer: (i) 1/16, (ii) 1/4, (iii) 11/16, (iv) 11/16.

9.5 If 20 percent of the bolts produced are defective, find the probability that in a sample of 4 bolts chosen at random, (i) none is defective, (ii) 2 are defective.

Step 1: Here n = 4, p = 0.2 and q = 0.8.

Step 2: (i) P(0 defective) = (0.8)^4 = 0.4096.

Step 3: (ii) P(2 defective) = C(4, 2) times (0.2)^2 times (0.8)^2 = 6 times 0.04 times 0.64 = 0.1536.

Answer: (i) 0.4096, (ii) 0.1536.

9.6 The probability of passing an examination is 0.75. Find the probability of (i) passing at least two examinations if you take six, (ii) failing at least two if you take four.

Step 1: (i) With n = 6 and p = 0.75, use the complement: P(at least 2 pass) = 1 - P(0 pass) - P(1 pass).

Step 2: P(0 pass) = (0.25)^6 = 0.000244 and P(1 pass) = 6 times 0.75 times (0.25)^5 = 0.004395.

Step 3: Hence P(at least 2 pass) = 1 - 0.000244 - 0.004395 = 0.995361, about 0.995.

Step 4: (ii) With n = 4, the probability of failing one examination is 0.25. P(at least 2 fail) = 1 - P(0 fail) - P(1 fail).

Step 5: P(0 fail) = (0.75)^4 = 0.316406 and P(1 fail) = 4 times 0.25 times (0.75)^3 = 0.421875.

Step 6: Hence P(at least 2 fail) = 1 - 0.316406 - 0.421875 = 0.261719, about 0.2617.

Answer: (i) 0.995, (ii) 0.2617.

9.7 A house agent provides satisfactory accommodation for 75 percent of his clients. If 6 clients approach him independently, find the probability that (i) less than 4 get satisfactory accommodation, (ii) at least 5 do.

Step 1: Here n = 6, p = 0.75 and q = 0.25.

Step 2: (i) P(less than 4) = P(0) + P(1) + P(2) + P(3).

Step 3: P(0) = (0.25)^6 = 0.000244; P(1) = 6 times 0.75 times (0.25)^5 = 0.004395; P(2) = 15 times (0.75)^2 times (0.25)^4 = 0.032959; P(3) = 20 times (0.75)^3 times (0.25)^3 = 0.131836.

Step 4: Adding them: 0.000244 + 0.004395 + 0.032959 + 0.131836 = 0.169434, about 0.1694.

Step 5: (ii) P(at least 5) = P(5) + P(6) = 6 times (0.75)^5 times 0.25 + (0.75)^6 = 0.355957 + 0.177979 = 0.533936, about 0.5340.

Answer: (i) 0.1694, (ii) 0.5340.

9.8 If there is a 20 percent chance that a worker will catch an occupational disease, find, out of 6 workmen, the probability that (i) not more than 2 catch the disease, (ii) 4 or more catch the disease.

Step 1: Here n = 6, p = 0.2 and q = 0.8. Writing probabilities with denominator 15625 (which is 5^6):

Step 2: P(0) = 4096/15625; P(1) = 6 times 0.2 times (0.8)^5 = 6144/15625; P(2) = 15 times (0.2)^2 times (0.8)^4 = 3840/15625.

Step 3: (i) P(not more than 2) = (4096 + 6144 + 3840) divided by 15625 = 14080/15625 = 0.90112.

Step 4: (ii) P(4 or more) = 1 - P(0) - P(1) - P(2) - P(3), and P(3) = 20 times (0.2)^3

times (0.8)^3 = 1280/15625.

Step 5: Hence P(4 or more) = (15625 - 14080 - 1280) divided by 15625 = 265/15625 = 0.01696.

Answer: (i) 14080/15625, (ii) 265/15625.

9.9 If 8 coins are tossed, find the probability of getting (i) exactly 5 heads, (ii) 1 to 7 heads.

Step 1: Here n = 8 and p = 1/2, so P(X = x) = C(8, x) divided by 256.

Step 2: (i) P(exactly 5 heads) = C(8, 5) divided by 256 = 56/256.

Step 3: (ii) P(1 to 7 heads) = 1 - P(0 heads) - P(8 heads) = 1 - 1/256 - 1/256 = 254/256.

Answer: (i) 56/256, (ii) 254/256.

9.10 If X is a binomial random variable with n = 10 and p = 0.4, find the mean and the variance of Y = (X - 10) divided by 6.

Step 1: For X, the mean is np = 10 times 0.4 = 4 and the variance is npq = 10 times 0.4 times 0.6 = 2.4.

Step 2: Y = (X - 10) divided by 6, so the mean of Y is (4 - 10) divided by 6 = -1.

Step 3: The variance of Y is the variance of X divided by 36, that is 2.4 divided by 36 = 0.0667.

Answer: Mean = -1, Variance = 0.067. Note: the question prints (X - 10) divided by 6, and the mean is negative for that reason; it is not a misprint to be corrected to (X - 4) divided by 6.

9.11 If X is the number of successes and p = 1/4 in 5 independent trials, find (i) P(X = 0), (ii) P(X at most 3).

Step 1: Here n = 5, p = 1/4 and q = 3/4, so P(X = x) = C(5, x) times (1/4)^x times (3/4)^(5-x).

Step 2: (i) P(X = 0) = (3/4)^5 = 243/1024.

Step 3: (ii) P(X at most 3) = 1 - P(4) - P(5) = 1 - 15/1024 - 1/1024 = 1008/1024.

Answer: (i) 243/1024, (ii) 1008/1024.

9.12 Five true dice are thrown once. Find the probability of getting 0, 1, 2, 3, 4 and 5 sixes, and the mean and variance of the distribution.

Step 1: Here n = 5 and p = 1/6, so P(X = x) = C(5, x) times (1/6)^x times (5/6)^(5-x), with denominator 7776.

Step 2: P(0) = 3125/7776; P(1) = 3125/7776; P(2) = 1250/7776; P(3) = 250/7776; P(4) = 25/7776; P(5) = 1/7776.

Step 3: Mean = np = 5 times 1/6 = 0.833.

Step 4: Variance = npq = 5 times 1/6 times 5/6 = 0.694. (The printed key gives 0.701; the value from npq is 0.694.)

Answer: Probabilities 3125/7776, 3125/7776, 1250/7776, 250/7776, 25/7776, 1/7776; Mean = 0.833, Variance = 0.694.

9.13 Five dice are tossed 96 times. Find the expected frequencies when throwing 4, 5 or 6 is a success.

Step 1: Success on one die has probability 3/6 = 1/2. For 5 dice, P(X = x) = C(5, x) divided by 32.

Step 2: The probabilities for x = 0, 1, 2, 3, 4, 5 are 1/32, 5/32, 10/32, 10/32, 5/32 and 1/32.

Step 3: Expected frequency = 96 times the probability, giving 3, 15, 30, 30, 15 and 3.

Answer: 3, 15, 30, 30, 15, 3.

9.14 Four dice are thrown and the number of sixes is recorded; the experiment is repeated 180 times. Find the theoretical frequencies of 0, 1, 2, 3 and 4 sixes, and the mean number of sixes in a single throw.

Step 1: Here n = 4 and p = 1/6, so P(X = x) = C(4, x) times (1/6)^x times (5/6)^(4-x).

Step 2: P(0) = 625/1296, P(1) = 500/1296, P(2) = 150/1296, P(3) = 20/1296, P(4) = 1/1296.

Step 3: Expected frequency = 180 times the probability: 180 times 625/1296 = 86.81, about 87; 180 times 500/1296 = 69.44, about 69; 180 times 150/1296 =

20.83, about 21; 180 times 20/1296 = 2.78, about 3; 180 times 1/1296 = 0.14, about 0.

Step 4: Mean number of sixes in a single throw = np = 4 times 1/6 = 0.67.

Answer: Frequencies 87, 69, 21, 3, 0; Mean = 0.67.

9.15 The mean of a binomial distribution is 6 and its variance is 2.4. Find p and n.

Step 1: Mean = np = 6 and variance = npq = 2.4.

Step 2: Dividing, q = variance divided by mean = 2.4 divided by 6 = 0.4, so p = 1 - 0.4 = 0.6.

Step 3: n = mean divided by p = 6 divided by 0.6 = 10.

Answer: p = 0.6, n = 10.

9.16 If the probability that a bolt is defective is 0.1 and 500 bolts are examined, find the mean and the standard deviation of the number of defective bolts.

Step 1: Here n = 500 and p = 0.1, so q = 0.9.

Step 2: Mean = np = 500 times 0.1 = 50.

Step 3: Variance = npq = 500 times 0.1 times 0.9 = 45, so the standard deviation is the square root of 45 = 6.71.

Answer: Mean = 50, S.D. = 6.71. Note: the printed key gives Mean = 40 and S.D. =

6, which are the values for n = 400, not n = 500; the value for the printed data (500 bolts) is 50 and 6.71.

9.17 The mean of a binomial distribution is 3 and its standard deviation is 1.5. Find the parameters of the distribution.

Step 1: Variance = (1.5)^2 = 2.25 and mean = np = 3.

Step 2: q = variance divided by mean = 2.25 divided by 3 = 0.75, so p = 0.25.

Step 3: n = mean divided by p = 3 divided by 0.25 = 12.

Answer: p = 0.25, n = 12.

9.18 Is it possible to have a binomial distribution with mean 5 and standard deviation 3.7? Give reasons.

No. For a binomial distribution the variance is npq and the mean is np, so variance divided by mean equals q, and q cannot exceed 1. Here the standard deviation 3.7 gives a variance of 13.69, which is greater than the mean 5, so q would be 13.69 divided by 5 = 2.738, which is impossible. Hence a binomial distribution with mean 5 and standard deviation 3.7 cannot exist.

9.19 For a binomial distribution with n = 5, if P(X = 0) = P(X = 1), find the mean of the distribution.

Step 1: P(X = 0) = q^5 and P(X = 1) = 5 times p times q^4.

Step 2: Setting them equal and dividing by q^4 gives q = 5p. Since p + q = 1, we get 6p = 1, so p = 1/6.

Step 3: Mean = np = 5 times 1/6 = 5/6.

Answer: p = 1/6, Mean = 5/6.

9.20 Discuss the statement that in a binomial distribution the mean is 6 and the standard deviation is 2.5.

No. The mean is 6 and the standard deviation is 2.5, so the variance is 6.25, which is greater than the mean. Then q = variance divided by mean = 6.25 divided by 6 = 1.042, which exceeds 1 and is impossible for a probability. Hence such a binomial distribution cannot exist.

9.21 A binomial distribution has mean 12 and standard deviation 3. Find the distribution.

Step 1: Variance = 9 and mean = np = 12, so q = 9 divided by 12 = 0.75 and p = 0.25.

Step 2: n = 12 divided by 0.25 = 48.

Step 3: The distribution is P(X = x) = C(48, x) times (0.25)^x times (0.75)^(48-x), for x = 0, 1, ..., 48.

Answer: p = 0.25, q = 0.75, n = 48, with P(X = x) as above.

9.22 Find the mean and variance of the binomial distribution (q + p)^3.

For (q + p)^3 the number of trials is n = 3, so the mean is np = 3p and the variance is npq = 3pq.

9.23 Find the mean and standard deviation of the binomial distribution (q + p)^n.

For (q + p)^n the mean is np, the variance is npq and the standard deviation is the

square root of npq.

Exercise 9 - 9.24 Fill in the Blanks

9.24 (i) The mean of a binomial distribution is _____ and its variance is _____.

The mean of a binomial distribution is np and its variance is npq.

9.24 (ii) The binomial distribution is symmetrical when p = q = 1/2 .

9.24 (iii) The binomial distribution is used when n is small .

9.24 (iv) The binomial distribution has two parameters.

9.24 (v) The binomial distribution is positively skewed when p < q .

9.24 (vi) The binomial variable is a discrete variable taking values X = 0, 1, 2, 3, ..., n.

9.24 (vii) The shape of the binomial distribution depends upon the values of _____.

The shape of the binomial distribution depends upon the values of n and p. (The book's key prints 'S.D.' here; the standard deviation, square root of npq, is itself fixed by n and p.)

9.24 (viii) A binomial experiment consists of a _____ number of _____ trials.

A binomial experiment consists of a fixed number of independent trials.

9.24 (ix) The mean, median and mode of a binomial distribution are equal when p = q .

Exercise 9 - Questions 9.25 to 9.29

9.25 Five balls are drawn from a box containing 4 white and 7 black balls. If X is the number of black balls drawn, find the distribution of X, its mean and variance, and verify the mean and variance by formula.

Step 1: The drawing is without replacement, so X follows a hypergeometric distribution. The total number of selections is C(11, 5) = 462.

Step 2: P(X = 1) = C(7, 1) times C(4, 4) divided by 462 = 7/462 = 1/66.

Step 3: P(X = 2) = C(7, 2) times C(4, 3) divided by 462 = 84/462 = 12/66.

Step 4: P(X = 3) = C(7, 3) times C(4, 2) divided by 462 = 210/462 = 30/66.

Step 5: P(X = 4) = C(7, 4) times C(4, 1) divided by 462 = 140/462 = 20/66.

Step 6: P(X = 5) = C(7, 5) times C(4, 0) divided by 462 = 21/462 = 3/66.

Step 7: Mean = (1 times 7 + 2 times 84 + 3 times 210 + 4 times 140 + 5 times 21) divided by 462 = 1470/462 = 35/11 = 3.1818. By formula, mean = n times K divided by N = 5 times 7 divided by 11 = 35/11.

Step 8: E(X^2) = (1 times 7 + 4 times 84 + 9 times 210 + 16 times 140 + 25 times 21) divided by 462 = 4998/462 = 10.8182, so variance = 10.8182 - (35/11)^2 = 0.6942. By formula, variance = n times (K/N) times (1 - K/N) times ((N - n)/(N - 1)) = 5 times (7/11) times (4/11) times (6/10) = 0.6942.

Answer: Distribution 1/66, 12/66, 30/66, 20/66, 3/66; Mean = 35/11; Variance = 0.6942, verified by formula.

9.26 A box contains 4 white and 7 black beads. Five beads are drawn at random. Find the distribution of the number of white beads drawn, its mean and variance, and check them by formula.

Step 1: The drawing is without replacement, so the number of white beads follows a hypergeometric distribution with N = 11, K = 4 white beads and n = 5 draws. Total

selections = C(11, 5) = 462.

Step 2: P(X = 0) = C(4, 0) times C(7, 5) divided by 462 = 21/462.

Step 3: P(X = 1) = C(4, 1) times C(7, 4) divided by 462 = 140/462.

Step 4: P(X = 2) = C(4, 2) times C(7, 3) divided by 462 = 210/462.

Step 5: P(X = 3) = C(4, 3) times C(7, 2) divided by 462 = 84/462.

Step 6: P(X = 4) = C(4, 4) times C(7, 1) divided by 462 = 7/462.

Step 7: Mean = (0 times 21 + 1 times 140 + 2 times 210 + 3 times 84 + 4 times 7) divided by 462 = 840/462 = 1.8181. By formula, mean = n times K divided by N = 5 times 4 divided by 11 = 1.8181.

Step 8: Variance = 0.6942, the same value as in Question 9.25 (white and black counts are complementary). By formula, variance = 5 times (4/11) times (7/11) times (6/10) = 0.6942.

Answer: Distribution 21/462, 140/462, 210/462, 84/462, 7/462; Mean = 1.8181; Variance = 0.6942, verified by formula.

9.27 A committee of 3 is to be formed from 4 men and 2 women. Find the hypergeometric distribution of the number of men on the committee.

Step 1: Total selections = C(6, 3) = 20.

Step 2: P(X = 1) = C(4, 1) times C(2, 2) divided by 20 = 4/20.

Step 3: P(X = 2) = C(4, 2) times C(2, 1) divided by 20 = 12/20.

Step 4: P(X = 3) = C(4, 3) times C(2, 0) divided by 20 = 4/20.

Answer: X = 1, 2, 3 with probabilities 4/20, 12/20, 4/20.

9.28 A committee of 5 is chosen from 3 women and 5 men. Find the expected number of women on the committee.

The committee has 5 members chosen from 3 women and 5 men. The expected number of women is 5 times 3 divided by 8 = 15/8 = 1.875.

9.29 Ten vegetable cans, 5 of tomatoes and 5 of corns, are placed in a box and 5 are selected at random. Find the probability that (i) all contain tomatoes, (ii) 3 or more contain tomatoes.

Step 1: Total selections = C(10, 5) = 252.

Step 2: (i) P(all 5 are tomatoes) = C(5, 5) divided by 252 = 1/252.

Step 3: (ii) P(3 or more tomatoes) = P(3) + P(4) + P(5) = (100 + 25 + 1) divided by 252 = 126/252.

Answer: (i) 1/252, (ii) 126/252.

Exercise 9 - 9.30 True or False

9.30 (i) An experiment is called a Bernoulli experiment if it has two possible

outcomes.

True

9.30 (ii) The binomial distribution has three parameters n, p and q.

False

9.30 (iii) A paper has 10 multiple-choice questions with 3 alternatives. Answering these questions by guesswork is a binomial experiment.

True

9.30 (iv) If p is not equal to q, then the binomial probability distribution is skewed.

True

9.30 (v) A binomial random variable is a continuous random variable.

False

9.30 (vi) The binomial distribution is a symmetrical distribution if p = q = 1/2.

True

9.30 (vii) The binomial distribution is a negatively skewed distribution if p > q.

True

9.30 (viii) The trials are independent in the hypergeometric distribution.

False

9.30 (ix) The hypergeometric probability distribution has three parameters n, N and k.

True

9.30 (x) The binomial distribution is used when 'n' is large.

False

9.30 (xi) The variance of the binomial distribution is 'npq'.

True

9.30 (xii) The binomial random variable cannot assume negative values.

True

9.30 (xiii) The binomial distribution is positively skewed distribution if p < q.

True

Exercise 9 - Note on the Answers key

Note on the book's printed Answers key (p. 261): the key marks 9.30 (i) F, (vii) F, (viii) T, (x) T.

The answers given here follow the chapter's definitions, and differ from the key at four places. (i) An experiment with exactly two possible outcomes is a Bernoulli experiment, so the statement is True. (vii) When p > q, that is p > 1/2, the binomial distribution is negatively skewed, so the statement is True. (viii) The hypergeometric distribution comes from sampling without replacement, so its trials are not independent, and the statement is False. (x) The binomial distribution applies for any fixed n; a large n is the condition for the normal approximation, not for the binomial distribution itself, so the statement is False. The four key entries are treated as misprints.

Note on 9.16: the printed key (p. 260) gives mean 40 and standard deviation 6. The question in 9.16 states a total of 500 bolts with defective probability 0.1, so the mean is 500 times 0.1 = 50 and the standard deviation is the square root of 500 times 0.1 times 0.9 = 6.71. The key's values 40 and 6 are the values for a total of 400, not 500, and are treated as misprints.